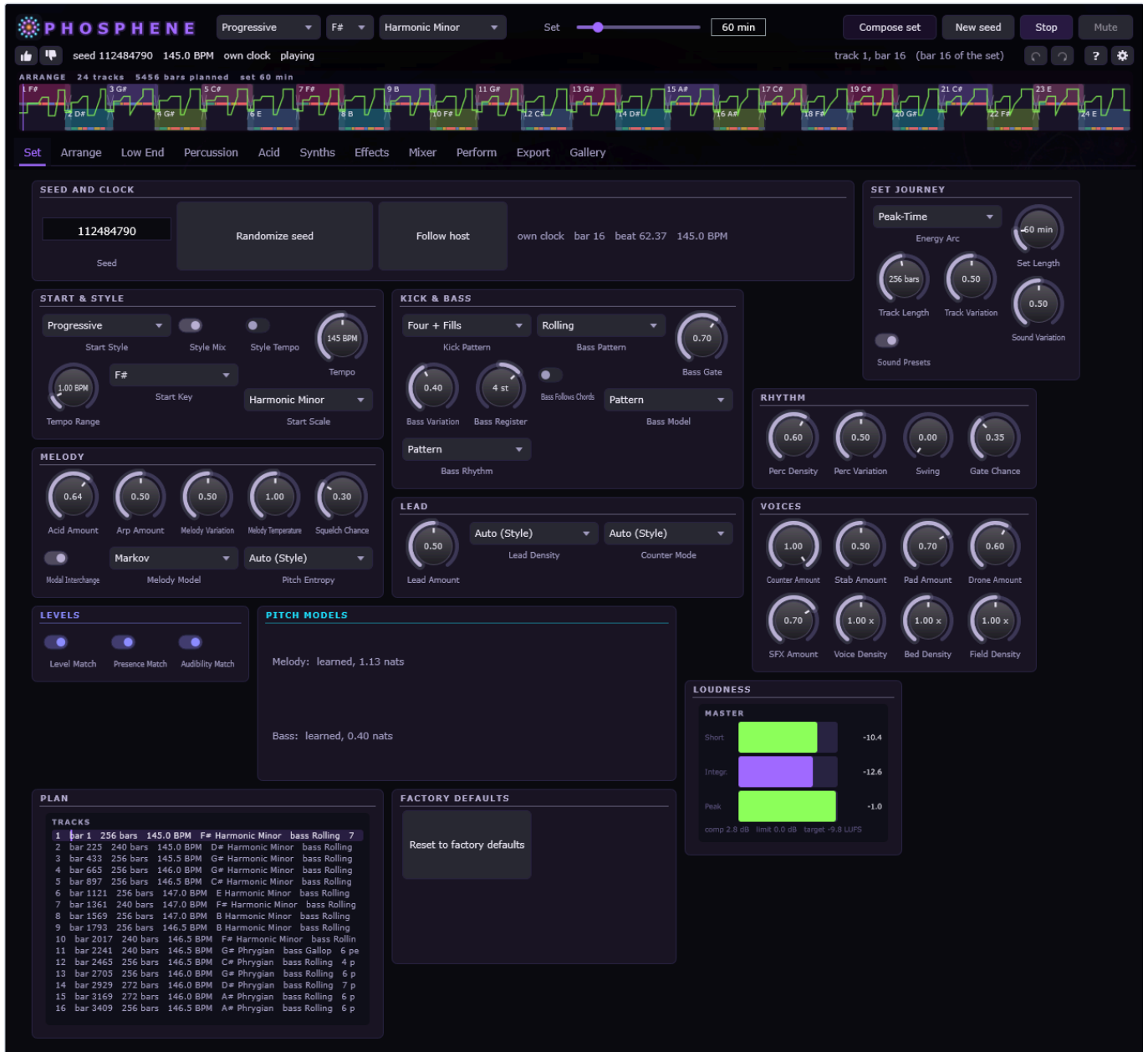


Phosphene

Manual · version 1.2.0



1552 parameters · 18 tabs · 4 perform macros · every picture in this manual is the plugin drawing itself at design size

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1. About

Phosphene composes and plays complete psytrance sets. You give it a seed, a style and an energy arc; it writes the tracks -- their keys, tempi, forms, patterns and sounds -- and synthesises them in real time. There is no loop pool and no sampled instrument: every note is decided by the composer, and every sound is made by the engine -- except the spoken phrases and, on the Field track, field recordings of real places (and of NASA's missions), which a place cannot be synthesised without.

Two ideas run through the whole instrument. The first is that the same seed always gives the same set: bar 3000 is bar 3000 whether the night was played from the start or joined in the middle, which is what makes a set something you can save in seven lines of text and hand to someone else. The second is that the composer never writes a knob. For every track it chooses a sound preset for each synth and plays it, and on top of that it rides offsets -- a section opening a filter, the level match evening out the parts. The knobs stay yours: the tab shows the preset the composer plays, a thin bright ring shows where a value plays away from its knob, and the moment you turn a knob the synth is yours again (Own Sound).

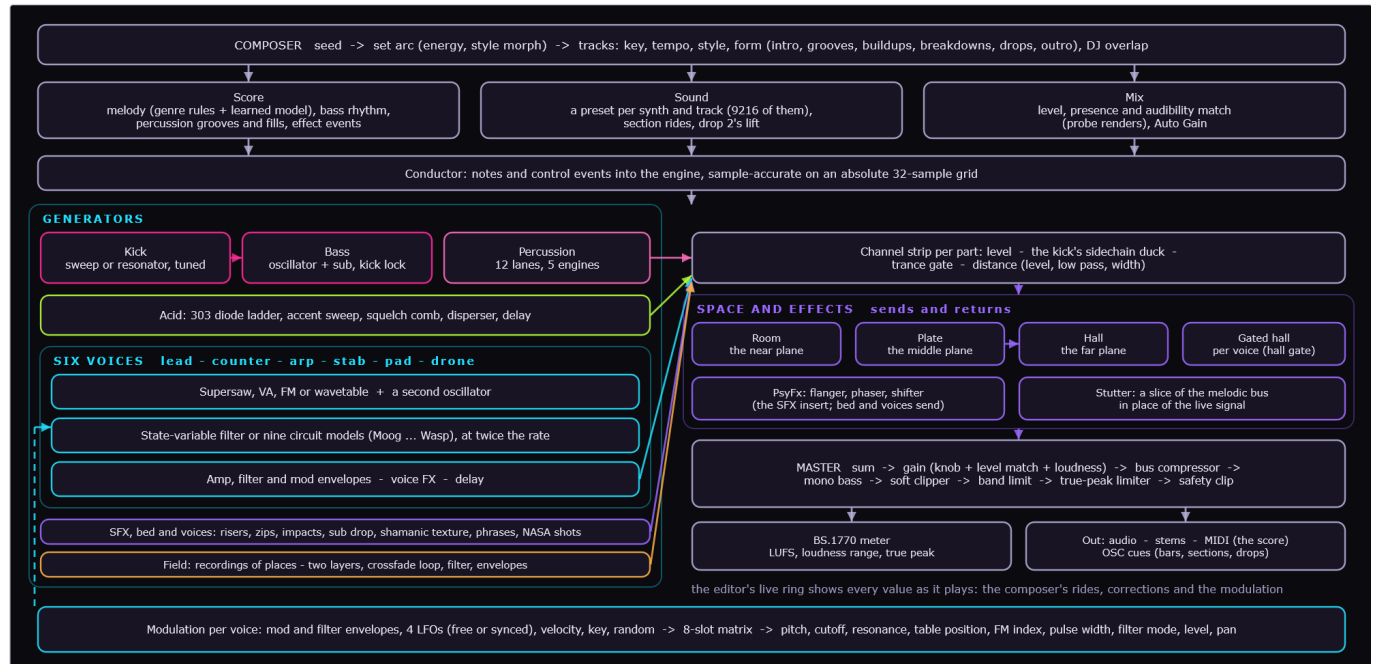
The panel has a tab per generator, plus the pages that are about the set rather than the sound: Set, Arrange, Mixer / Master, Perform and Gallery. Every control on every page is generated from the engine's own parameter tables, and so is this manual.

2. Signal flow

The picture is the whole instrument in the order it works. At the top the composer: a seed becomes a set arc, the arc becomes tracks with keys, tempi, styles and forms, and each track becomes a score, a choice of sounds and a mix -- the last measured by rendering parts of the track before it plays. The conductor hands notes and control events to the engine on an absolute grid of 32 samples, which is why a bar sounds the same however a host cuts its blocks.

On the left the generators, each in the colour of its tab: kick, bass and percussion, the acid, the six melodic voices with their oscillators, filter models and envelopes, the effects with the bed and the spoken voices, and the Field track's sampler. Every part has a channel strip -- level, the kick's sidechain duck, the trance gate and its distance -- and sends into three planes of space: a room near, a plate in the middle, a hall far behind, plus a gated hall a voice can choose. The master sums it all and brings it to loudness: gain, bus compressor, mono bass, soft clipper, band limit, a true-peak limiter and a safety clip, metered to BS.1770. Out go the audio, the stems, the score as MIDI and the cues over OSC.

Along the bottom the modulation of every voice: its envelopes, four LFOs and an eight-slot matrix.



The signal flow, drawn by the plugin (the same picture as the help page's).

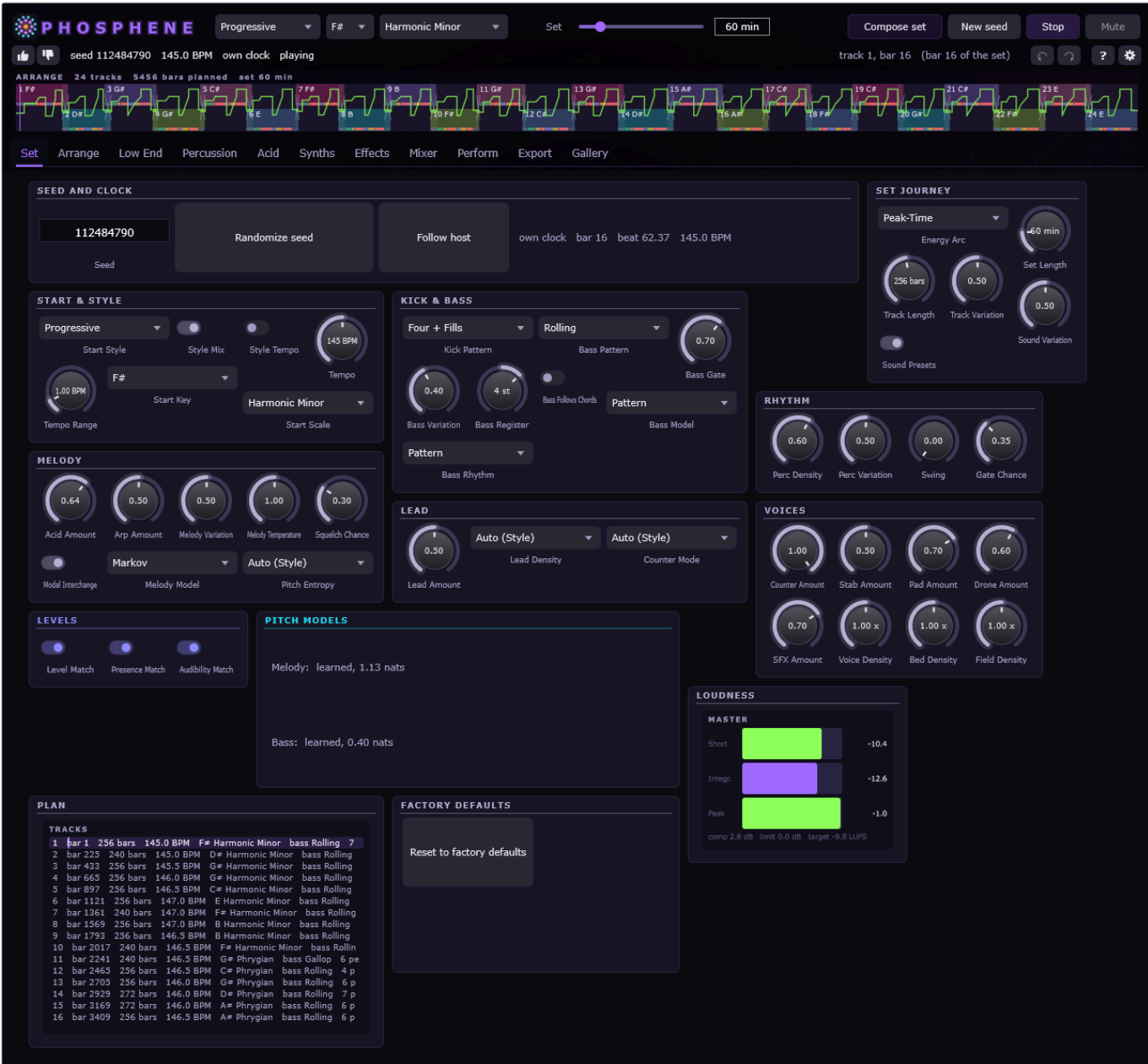
3. Set

The Set tab is where a night begins. The seed names the set; Randomize seed draws another one. Play and Stop run Phosphene's own clock, and Follow host hands the transport to the DAW instead -- under a host the tempo knob follows the host's tempo and the play head follows its position.

Planning a track measures it by rendering parts of it -- its loudness through the master, the brightness of its lines, how audible each line is against the lead. For the first track of a set that happens behind the music: it starts playing at once, its intro runs, and the measured levels ride in over a phrase long before the kick and the lines arrive. A jump to a track that has not been planned yet still waits for its measurements; the header says "planning" rather than going quiet without explanation. In the standalone this has usually happened before you press anything, because the composer thread starts working when the window opens.

The composer's own knobs are here, grouped by what they decide. Start & Style is where the night begins: the style profile, whether the style walks through the set (Style Mix) and whether each track takes its style's own tempo (Style Tempo, which then replaces Tempo and Tempo Range), and the start key and scale. They are called "start" because they are exactly that -- the first track plays them, and from the second on the key, the tempo and (with Style Mix) the style move from track to track as a DJ would move them. Set Journey is the shape of the night: the energy arc, the set length, the track length and how much tracks and sounds vary. Kick & Bass, Rhythm, Melody, Lead and Voices say how much of each part there is and how it is drawn; Levels holds the three automatic balances -- the loudness of whole tracks, the presence of the melodic lines, and the audibility of the quiet lines against the lead.

Export MIDI writes the score of the first N bars as a Standard MIDI File with a named track per part, section markers and the tempo map. Export set writes a .phosset - the seed, the style, the arc, the knobs that differ from their defaults, and every lock and reroll -- which phos_render will play back byte for byte.



The Set tab.

START & STYLE

Control	Key	Range	Default
Start Style	compose.style	Goa Full-On Progressive Dark Forest Hi-Tech	Full-On

Style Mix	compose.style_mix	off / on	on
Style Tempo	compose.style_tempo	off / on	off
Tempo	compose.bpm	100 .. 190 BPM	145
Tempo Range	compose.tempo_range	0 .. 10 BPM	1
Start Key	compose.key	C C# D D# E F F# G G# A A# B	F#
Start Scale	compose.scale	Aeolian Phrygian Harmonic Minor Phrygian Dominant Double Harmonic Dorian	Phrygian

SET JOURNEY

Control	Key	Range	Default
Energy Arc	compose.arc	Warm-up Peak-Time Morning Closing Flat	Flat
Set Length	compose.set_minutes	10 .. 300 min	60
Track Length	compose.track_bars	32 .. 512 bars	256
Track Variation	compose.track_variation	0 .. 1	0.5
Sound Variation	compose.sound_variation	0 .. 1	0.5
Sound Presets	compose.sound_presets	off / on	on

KICK & BASS

Control	Key	Range	Default
Kick Pattern	compose.kick_pattern	Four Four + Fills Off	Four + Fills
Bass Pattern	compose.bass_pattern	Rolling Gallop Skip Offbeat Triplet	Rolling
Bass Gate	compose.bass_gate	0.2 .. 1	0.7
Bass Variation	compose.bass_variation	0 .. 1	0.4
Bass Register	compose.bass_register	-12 .. 12 st	0
Bass Follows Chords	compose.bass_follows_chords	off / on	off
Bass Model	compose.bass_model	Pattern Neural	Pattern
Bass Rhythm	compose.bass_rhythm	Pattern Corpus	Pattern

RHYTHM

Control	Key	Range	Default
Perc Density	compose.perc_density	0 .. 1	0.6
Perc Variation	compose.perc_variation	0 .. 1	0.5
Swing	compose.swing	0 .. 0.3	0
Gate Chance	compose.gate_chance	0 .. 1	0.35

MELODY

Control	Key	Range	Default
Acid Amount	compose.acid_amount	0 .. 1	0.6
Arp Amount	compose.arp_amount	0 .. 1	0.5
Melody Variation	compose.melody_variation	0 .. 1	0.5
Melody Temperature	compose.melody_temperature	0.5 .. 2	1
Squelch Chance	compose.squelch_chance	0 .. 1	0.3
Modal Interchange	compose.modal_interchange	off / on	on
Melody Model	compose.melody_model	Markov Neural	Markov
Pitch Entropy	compose.pitch_entropy	Auto (Style) Low Medium High	Auto (Style)

LEAD

Control	Key	Range	Default
Lead Amount	compose.lead_amount	0 .. 1	0.5
Lead Density	compose.lead_density	Auto (Style) Sparse Medium Dense	Auto (Style)
Counter Mode	compose.counter_mode	Auto (Style) Echo Answer Timbral Hocket	Auto (Style)

VOICES

Control	Key	Range	Default
Counter Amount	compose.counter_amount	0 .. 1	1
Stab Amount	compose.stab_amount	0 .. 1	0.5
Pad Amount	compose.pad_amount	0 .. 1	0.7
Drone Amount	compose.drone_amount	0 .. 1	0.6
SFX Amount	compose.sfx_amount	0 .. 1	0.7

Voice Density	compose.voice_density	0 .. 2 x	1
Bed Density	compose.bed_density	0 .. 2 x	1
Field Density	compose.field_density	0 .. 2 x	1

LEVELS

Control	Key	Range	Default
Level Match	compose.level_match	off / on	on
Presence Match	compose.presence_match	off / on	on
Audibility Match	compose.audibility_match	off / on	on

4. Arrange

The Arrange tab is the whole planned set on one page, and the curation loop that goes with it.

The strip across the top is the night: every planned track a block in the order it plays, the energy arc drawn over it, the play head in it. Underneath, each track gets a row of its own in which its own form fills the full width, so a section stays readable no matter how long the set is. The colours are the section categories: intro and outro grey, groove green, buildup amber, the pre-drop break violet, the drop red, the breakdown blue. Clicking any block jumps there.

The two small glyphs on every track and every section are the answer to the oldest complaint about generators, which is that they are boring. The padlock freezes that unit on the seed it has; the die draws it again. Because a locked unit keeps its old seed and nothing later reads anything an earlier one decided, rerolling one track leaves every other track bit-identical -- so you can keep what you like and reroll what you do not, one section at a time, until the set is yours. Reroll the set draws the whole journey again and leaves the locked tracks exactly where they were.



The Arrange tab.

5. Kick

The kick is the floor of the genre and the first thing to get right. Two engines: a swept sine whose pitch falls from Pitch Start to Pitch End with its own decay, and a damped phasor resonator with the retrigger behaviour of a TR-808. Punch is a second, much faster pitch envelope on top of the first; Click adds the attack transient; Drive and Clip decide how much of the body survives.

Two of its numbers are not free. The tail is limited so that the kick has fallen far enough by the time the first bass note starts, and the phase lock (on the Bass tab) can trim the kick's decay so that the kick and the bass fundamental meet in phase. Both are applied in the engine, after every knob and every variation.

The Kick Scope renders one hit with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, its pitch sweep on a log axis, its spectrum, and in figures where the sweep starts, the note it lands on (with its cents off it), its length to -60 dB and its peak.



The Kick tab.

SOUND

Control	Key	Range	Default
Kick Own Sound	<code>mix.kick_own</code>	off / on	off

PITCH

Control	Key	Range	Default
Engine	<code>kick.engine</code>	Sweep Resonant	Sweep
Tune	<code>kick.tune</code>	Free Key	Key
Pitch Start	<code>kick.pitch_start</code>	60 .. 1500 Hz	330
Pitch End	<code>kick.pitch_end</code>	30 .. 120 Hz	50
Body Decay	<code>kick.pitch_decay</code>	5 .. 150 ms	13

PUNCH & BODY

Control	Key	Range	Default
Punch	kick.punch	0 .. 1	0.5
Punch Decay	kick.punch_decay	0.5 .. 20 ms	4
Decay	kick.amp_decay	20 .. 1500 ms	240
Attack	kick.amp_attack	0 .. 10 ms	0.2
Hold	kick.amp_hold	0 .. 150 ms	12

CLICK

Control	Key	Range	Default
Click	kick.click_level	0 .. 1	0.5
Click Tone	kick.click_tone	500 .. 12000 Hz	4000
Click Decay	kick.click_decay	0.5 .. 30 ms	3

DRIVE & OUTPUT

Control	Key	Range	Default
Level	kick.level	-36 .. 6 dB	-6
Drive	kick.drive	0 .. 1	0.3
Clip	kick.clip	Tanh Hard	Tanh
Tone	kick.tone	200 .. 20000 Hz	9000
Tail Limit	kick.tail_limit	-60 .. 0 dB	-15

6. Bass

The rolling bass and the kick are one instrument with two halves. The oscillator is a PolyBLEP saw and pulse with a sine sub on the fundamental -- on the fundamental, not an octave down, because the octave-down sub fought the kick -- and the filter is a non-linear zero-delay ladder.

Kick Lock decides which of the two gives way: Bass follows kick sets the bass start phase to the kick's phase at that instant, Kick follows bass trims the kick's decay by bisection until the two agree. Off leaves them where they fall, which at 145 BPM means anything between -157 and +165 degrees.

Pattern and Gate are the note side: which of the five bass patterns the track prefers, and how long a note is as a fraction of its slot. The gate is never allowed to run shorter than half a period of the fundamental.

The bass is a pedal on the root. Its figures at the phrase ends may touch the b2 or the leading tone only as a pickup -- at most two sixteenths, at the end of the bar, into the next downbeat; anywhere else the figure takes the fifth, because under the drone's tonic those notes are a minor ninth in the low register. Bass Follows Chords (Set tab) moves the bass to the root the pad holds, and only where the pad plays; the drone moves with it.

The Bass Scope renders one eighth note of the bass on the key's root with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, the filter's cutoff over the note as the synth computes it, its spectrum, and the note and the filter's travel in figures.



The Bass tab.

SOUND

Control	Key	Range	Default
Bass Own Sound	<code>mix.bass_own</code>	off / on	off

OSCILLATOR

Control	Key	Range	Default
Wave	<code>bass.wave</code>	0 .. 1	0.15
Sub Mode	<code>bass.sub_mode</code>	Mixed Split	Split

Pulse Width	bass.pulse_width	0.05 .. 0.95	0.5
Sub	bass.sub	0 .. 1	0.3
Sub Octave	bass.sub_octave	0 .. 1	0.5
Split	bass.split_ratio	1.2 .. 3 x f0	2
Start Phase	bass.start_phase	0 .. 1	0.5
Retrigger	bass.retrigger	off / on	on
Kick Lock	bass.kick_lock	Off Bass follows kick Kick follows bass	Kick follows bass

FILTER

Control	Key	Range	Default
Filter Model	bass.filter_model	Bass Ladder Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	Bass Ladder
Cutoff	bass.cutoff	20 .. 10000 Hz	240
Resonance	bass.resonance	0 .. 1	0.3
Env Amount	bass.env_amount	0 .. 8 oct	4
Filter Mode	bass.filter_mode	0 .. 1	0
Key Track	bass.key_track	0 .. 1	0.6
Vel > Cutoff	bass.vel_to_cutoff	0 .. 1	0.25

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	bass.filt_attack	0.1 .. 4000 ms	0.1
Filter Decay	bass.filter_decay	3 .. 1000 ms	75
Filter Sustain	bass.filt_sustain	0 .. 1	0
Filter Release	bass.filt_release	5 .. 8000 ms	300

BITE

Control	Key	Range	Default
Bite	bass.bite	0 .. 1	0.5
Bite Cutoff	bass.bite_cutoff	100 .. 4000 Hz	500
Bite Reso	bass.bite_resonance	0 .. 1	0.2
Bite Env	bass.bite_env	0 .. 5 oct	3
Bite Decay	bass.bite_decay	5 .. 1000 ms	90
Bite Drive	bass.bite_drive	0 .. 1	0.5

AMPLIFIER

Control	Key	Range	Default
Drive	bass.drive	0 .. 1	0.3
Attack	bass.amp_attack	0.1 .. 50 ms	0.8
Decay	bass.amp_decay	5 .. 2000 ms	180
Sustain	bass.amp_sustain	0 .. 1	0.55
Release	bass.amp_release	1 .. 500 ms	10

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	bass.menv_attack	0.1 .. 8000 ms	10
Mod Decay	bass.menv_decay	5 .. 12000 ms	800
Mod Sustain	bass.menv_sustain	0 .. 1	0
Mod Release	bass.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	bass.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	bass.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	bass.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrig	bass.lfo1_retrig	off / on	off
LFO 1 Fade	bass.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	bass.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	bass.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	bass.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	bass.lfo2_retrigger	off / on	off
LFO 2 Fade	bass.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	bass.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	bass.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	bass.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	bass.lfo3_retrigger	off / on	off
LFO 3 Fade	bass.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	bass.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	bass.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	bass.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	bass.lfo4_retrigger	off / on	off
LFO 4 Fade	bass.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	bass.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	bass.mx1_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 1 Amount	bass.mx1_amount	-1 .. 1	0
Mod 2 Source	bass.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	bass.mx2_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 2 Amount	bass.mx2_amount	-1 .. 1	0
Mod 3 Source	bass.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	bass.mx3_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 3 Amount	bass.mx3_amount	-1 .. 1	0
Mod 4 Source	bass.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	bass.mx4_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 4 Amount	bass.mx4_amount	-1 .. 1	0
Mod 5 Source	bass.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	bass.mx5_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 5 Amount	bass.mx5_amount	-1 .. 1	0
Mod 6 Source	bass.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	bass.mx6_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 6 Amount	bass.mx6_amount	-1 .. 1	0
Mod 7 Source	bass.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	bass.mx7_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 7 Amount	bass.mx7_amount	-1 .. 1	0
Mod 8 Source	bass.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	bass.mx8_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 8 Amount	bass.mx8_amount	-1 .. 1	0

DUCK & LEVEL

Control	Key	Range	Default
Level	bass.level	-36 .. 6 dB	-3
Duck Depth	bass.duck_depth	0 .. 1	0.37
Duck Hold	bass.duck_hold	0 .. 200 ms	25
Duck Release	bass.duck_release	5 .. 500 ms	60

7. Percussion

Twelve lanes, each a complete voice: a tone with its own pitch envelope and FM, four modal sets, the 808 square stack, noise with clap bursts, a filter, a low cut, drive, choke and panning. The role of a lane decides what it plays in the groove and which General MIDI note it exports as.

The patterns are euclidean, rotated to a middle amount of syncopation rather than to none, and the decisions are made per four-bar phrase. Every lane is high-passed at 150 Hz at the least: the bottom of the mix belongs to the kick and the bass, and a tom that reaches into it takes the floor away.

On a metal lane, Noise trades against the square stack: at 1 the squares keep 20 % of their level. Six summed squares are six steady tones, and a cymbal made mostly of them rings like feedback. The default hats and ride therefore run almost all noise, with their low cuts at 3 .. 3.5 kHz.



The Percussion tab.

The 12 lanes of the kit share one table; lane 1 is printed here and the others are the same controls under the keys perc2.* to perc12.*.

LANE

Control	Key	Range	Default
Active	perc1.active	off / on	on
Role	perc1.role	Closed Hat Open Hat Ride Crash Clap Snare Rim Shaker Tom Conga Zap Blip	Closed Hat
Engine	perc1.engine	Noise Metal Modal Tone FM	Metal
Density	perc1.density	0 .. 1	0.6
Choke Group	perc1.choke	0 .. 4	1
Cut Tracks Pitch	perc1.cut_track	0 .. 2	0

SOURCE

Control	Key	Range	Default
Pitch	perc1.pitch	40 .. 12000 Hz	400
Pitch Amount	perc1.pitch_amount	1 .. 16 x	1
Pitch Decay	perc1.pitch_decay	0.5 .. 300 ms	10
Tune to Key	perc1.tune	off / on	off
Shift	perc1.shift	-10 .. 10 ms	0
FM Ratio	perc1.fm_ratio	0.25 .. 8	1.41
FM Index	perc1.fm_index	0 .. 8	0

Modes	perc1.mode_set	Membrane Bar Harmonic	Membrane
Mode Damping	perc1.mode_damp	0 .. 1	0.5
Metal Scale	perc1.metal_scale	0.25 .. 4 x	1

NOISE

Control	Key	Range	Default
Noise	perc1.noise	0 .. 1	1
Noise Decay	perc1.noise_decay	2 .. 3000 ms	35
Bursts	perc1.bursts	1 .. 6	1
Burst Spacing	perc1.burst_spacing	2 .. 40 ms	10

FILTER & SHAPE

Control	Key	Range	Default
Filter	perc1.filter	Low Pass Band Pass High Pass	Low Pass
Cutoff	perc1.cutoff	100 .. 18000 Hz	12000
Resonance	perc1.resonance	0 .. 1	0.05
Low Cut	perc1.low_cut	150 .. 8000 Hz	3000
Decay	perc1.decay	2 .. 3000 ms	45
Drive	perc1.drive	0 .. 1	0

OUTPUT & MOTION

Control	Key	Range	Default
Level	perc1.level	-36 .. 6 dB	1.5
Pan	perc1.pan	-1 .. 1	0.45
Pan Depth	perc1.pan_depth	0 .. 1	1
Pan Period	perc1.panBars	0.0625 .. 16 bars	0.1875

8. Acid

A diode ladder after Zavalishin, solved from the inside out, with an accent that charges a sweep capacitor the way the original did -- three accents in a row open further than one. Slide is an exponential glide into the next note, Squelch a cutoff start value plus a comb tuned to the note's period. The tempo delay is in beats, so it follows the set.

In the Phrygian modes the b2 is a degree of the acid's own: the F of an E - F - E riff, on any step and held too. The other colour tones -- the upper note of an augmented second -- stay short neighbours that resolve to the tonic.



The Acid tab.

SOUND

Control	Key	Range	Default
Acid Own Sound	<code>mix.acid_own</code>	off / on	off

VOICE

Control	Key	Range	Default
Wave	<code>acid.wave</code>	0 .. 1	0
Slide Time	<code>acid.slide_time</code>	5 .. 200 ms	70
Amp Decay	<code>acid.amp_decay</code>	50 .. 4000 ms	900

FILTER

Control	Key	Range	Default
Cutoff	acid.cutoff	80 .. 8000 Hz	900
Resonance	acid.resonance	0 .. 1	0.72
Env Amount	acid.env_amount	0 .. 6 oct	3.5
Decay	acid.decay	30 .. 2000 ms	220
Accent	acid.accent	0 .. 1	0.7
Key Track	acid.key_track	0 .. 1	0.5

SQUELCH & DRIVE

Control	Key	Range	Default
Squelch	acid.squelch	off / on	off
Drive	acid.drive	0 .. 1	0.85
Squelch Start	acid.squelch_start	2 .. 32 x	12
Squelch Time	acid.squelch_time	5 .. 150 ms	28
Low Cut	acid.low_cut	150 .. 1000 Hz	250
Comb Mix	acid.comb_mix	0 .. 1	0.55
Comb Feedback	acid.comb_feedback	0 .. 0.97	0.82
Disperse	acid.disperse	0 .. 8 x	0
Disperse Freq	acid.disperse_freq	200 .. 8000 Hz	1250

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	acid.menv_attack	0.1 .. 8000 ms	10
Mod Decay	acid.menv_decay	5 .. 12000 ms	800
Mod Sustain	acid.menv_sustain	0 .. 1	0
Mod Release	acid.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	acid.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	acid.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	acid.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	acid.lfo1_retrigger	off / on	off
LFO 1 Fade	acid.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	acid.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	acid.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	acid.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	acid.lfo2_retrigger	off / on	off
LFO 2 Fade	acid.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	acid.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	acid.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	acid.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	acid.lfo3_retrigger	off / on	off
LFO 3 Fade	acid.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	acid.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	acid.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	acid.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	acid.lfo4_retrigger	off / on	off

LFO 4 Fade	acid.lfo4_fade	0 .. 8 s	0
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MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	acid.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	acid.mx1_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 1 Amount	acid.mx1_amount	-1 .. 1	0
Mod 2 Source	acid.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	acid.mx2_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 2 Amount	acid.mx2_amount	-1 .. 1	0
Mod 3 Source	acid.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	acid.mx3_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 3 Amount	acid.mx3_amount	-1 .. 1	0
Mod 4 Source	acid.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	acid.mx4_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 4 Amount	acid.mx4_amount	-1 .. 1	0
Mod 5 Source	acid.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	acid.mx5_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 5 Amount	acid.mx5_amount	-1 .. 1	0
Mod 6 Source	acid.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	acid.mx6_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 6 Amount	acid.mx6_amount	-1 .. 1	0
Mod 7 Source	acid.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	acid.mx7_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 7 Amount	acid.mx7_amount	-1 .. 1	0
Mod 8 Source	acid.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	acid.mx8_dst	Off Pitch Cutoff Resonance Level Pan	Off
Mod 8 Amount	acid.mx8_amount	-1 .. 1	0

DELAY

Control	Key	Range	Default
Delay Send	acid.delay_send	0 .. 1	0.25
Delay Feedback	acid.delay_feedback	0 .. 0.9	0.5
Delay Left	acid.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	acid.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay High Pass	acid.delay_high_pass	150 .. 2000 Hz	400
Delay Low Pass	acid.delay_low_pass	800 .. 16000 Hz	4500

SPACE & OUTPUT

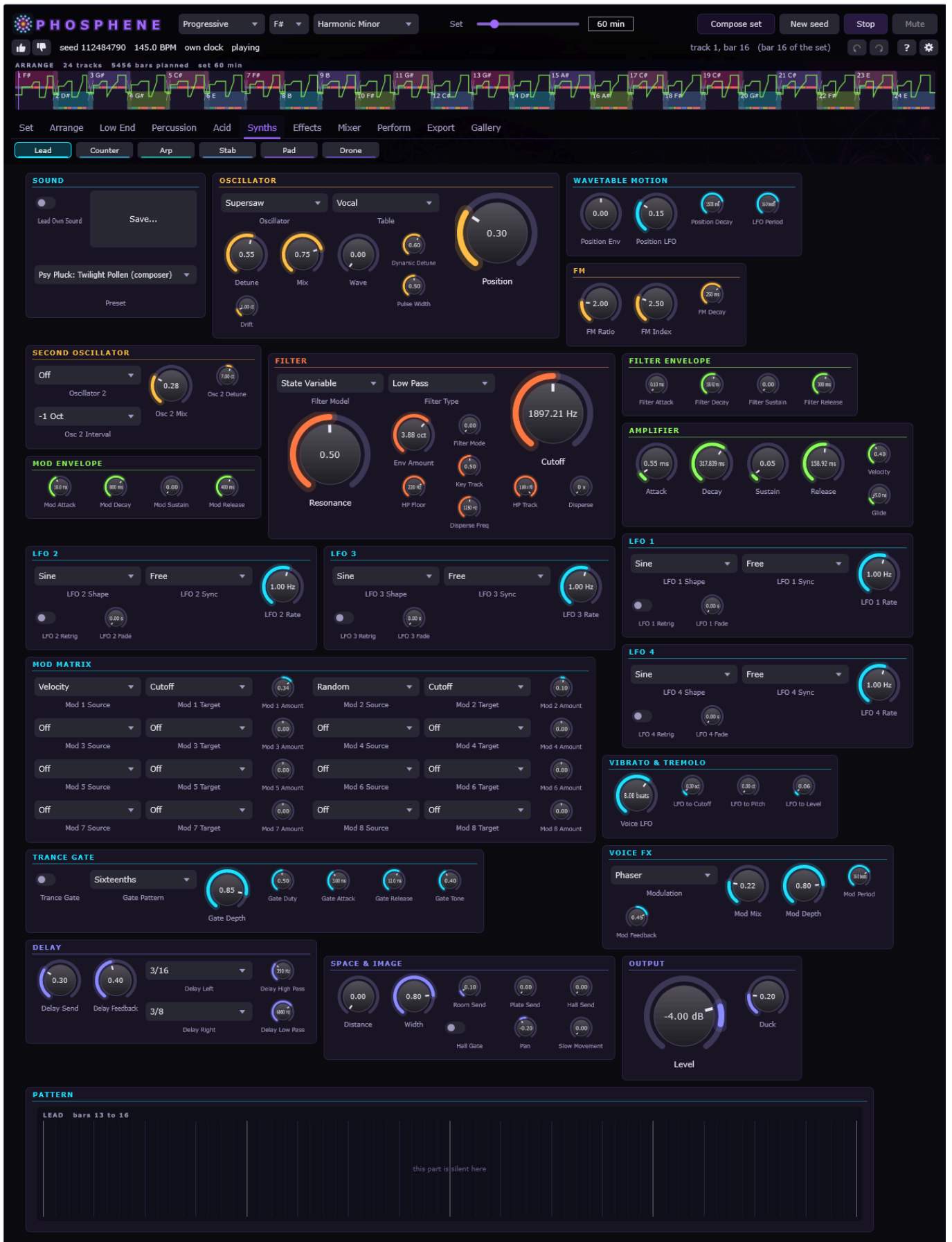
Control	Key	Range	Default
Level	acid.level	-36 .. 6 dB	-2.3
Duck	acid.duck	0 .. 1	0.15
Room Send	acid.room_send	0 .. 1	0.1
Plate Send	acid.plate_send	0 .. 1	0.15
Hall Send	acid.hall_send	0 .. 1	0
Hall Gate	acid.hall_gate	off / on	off

9. Lead

A supersaw after Szabo's measurements of the JP-8000 -- the detune curve, the mix law and the random phase per note -- with a dynamic detune that follows the length of the note. The other three oscillator types are a virtual-analogue saw and pulse, two-operator FM and a wavetable.

Everything but the kick and the bass is high-passed at its own fundamental, which is what keeps eight voices of supersaw out of the bass register.

As in the acid, the Phrygian b2 is one of the lead's degrees, on beats 2 and 4 and off the beat; beats 1 and 3 take a tone of the tonic chord.



The Lead tab.

SOUND

Control	Key	Range	Default
Lead Own Sound	<code>mix.lead_own</code>	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	lead.osc	Supersaw VA FM Wavetable
Table	lead.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo WaveEd
Position	lead.position	0 .. 1
Detune	lead.detune	0 .. 1
Mix	lead.mix	0 .. 1
Wave	lead.wave	0 .. 1
Dynamic Detune	lead.dynamic_detune	0 .. 1
Pulse Width	lead.pulse_width	0.05 .. 0.95
Drift	lead.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	lead.pos_env	-1 .. 1	0
Position LFO	lead.pos_lfo_depth	0 .. 0.5	0.15
Position Decay	lead.pos_decay	10 .. 8000 ms	1500
LFO Period	lead.pos_lfo_beats	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	lead.fm_ratio	0.5 .. 8	2
FM Index	lead.fm_index	0 .. 10	2.5
FM Decay	lead.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	lead.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	lead.osc2_mix	0 .. 1	0.28
Osc 2 Interval	lead.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	lead.osc2_detune	-50 .. 50 ct	7

FILTER

Control	Key	Range	Default
Filter Model	lead.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	lead.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	lead.cutoff	200 .. 18000 Hz	10000
Resonance	lead.resonance	0 .. 1	0.15
Env Amount	lead.env_amount	0 .. 6 oct	2
Filter Mode	lead.filter_mode	0 .. 1	0
Key Track	lead.key_track	0 .. 1	0.5
HP Floor	lead.hp_floor	40 .. 400 Hz	220
HP Track	lead.hp_track	0 .. 1 x f0	1
Disperse	lead.disperse	0 .. 8 x	0
Disperse Freq	lead.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	lead.filt_attack	0.1 .. 4000 ms	0.1
Filter Decay	lead.filter_decay	5 .. 3000 ms	400
Filter Sustain	lead.filt_sustain	0 .. 1	0
Filter Release	lead.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	lead.amp_attack	0.3 .. 2000 ms	4
Decay	lead.amp_decay	5 .. 4000 ms	500
Sustain	lead.amp_sustain	0 .. 1	0.75
Release	lead.amp_release	5 .. 4000 ms	180
Velocity	lead.vel_sens	0 .. 1	0.4
Glide	lead.glide	0 .. 400 ms	45

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	lead.menv_attack	0.1 .. 8000 ms	10
Mod Decay	lead.menv_decay	5 .. 12000 ms	800
Mod Sustain	lead.menv_sustain	0 .. 1	0
Mod Release	lead.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	lead.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	lead.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	lead.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	lead.lfo1_retrigger	off / on	off
LFO 1 Fade	lead.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	lead.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	lead.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	lead.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	lead.lfo2_retrigger	off / on	off
LFO 2 Fade	lead.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	lead.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	lead.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	lead.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	lead.lfo3_retrigger	off / on	off
LFO 3 Fade	lead.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	lead.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	lead.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	lead.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	lead.lfo4_retrigger	off / on	off
LFO 4 Fade	lead.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	lead.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	lead.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	lead.mx1_amount	-1 .. 1	0
Mod 2 Source	lead.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	lead.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	lead.mx2_amount	-1 .. 1	0
Mod 3 Source	lead.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	lead.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	lead.mx3_amount	-1 .. 1	0

Mod 4 Source	lead.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	lead.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	lead.mx4_amount	-1 .. 1	0
Mod 5 Source	lead.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	lead.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	lead.mx5_amount	-1 .. 1	0
Mod 6 Source	lead.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	lead.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	lead.mx6_amount	-1 .. 1	0
Mod 7 Source	lead.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	lead.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	lead.mx7_amount	-1 .. 1	0
Mod 8 Source	lead.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	lead.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	lead.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	lead.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	lead.lfo_cutoff	0 .. 4 oct	0.3
LFO to Pitch	lead.lfo_pitch	0 .. 50 ct	0
LFO to Level	lead.lfo_amp	0 .. 1	0.06

VOICE FX

Control	Key	Range	Default
Modulation	lead.mod	Off Flanger Phaser Comb	Phaser
Mod Mix	lead.mod_mix	0 .. 1	0.22
Mod Depth	lead.mod_depth	0 .. 1	0.8
Mod Period	lead.mod_beats	0.5 .. 64 beats	16
Mod Feedback	lead.mod_feedback	-0.9 .. 0.9	0.45

TRANCE GATE

Control	Key	Range	Default
Trance Gate	lead.gate	off / on	off
Gate Pattern	lead.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	lead.gate_depth	0 .. 1	0.85
Gate Duty	lead.gate_duty	0.05 .. 1	0.5
Gate Attack	lead.gate_attack	0.5 .. 60 ms	3
Gate Release	lead.gate_release	0.5 .. 120 ms	12
Gate Tone	lead.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	lead.delay_send	0 .. 1	0.3
Delay Feedback	lead.delay_feedback	0 .. 0.9	0.4
Delay Left	lead.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	lead.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	3/8
Delay High Pass	lead.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	lead.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	lead.distance	0 .. 1	0
Width	lead.width	0 .. 1	0.8
Room Send	lead.room_send	0 .. 1	0.1
Plate Send	lead.plate_send	0 .. 1	0
Hall Send	lead.hall_send	0 .. 1	0
Hall Gate	lead.hall_gate	off / on	off

Pan	lead.pan	-1 .. 1	-0.2
Slow Movement	lead.slow_mod	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	lead.level	-36 .. 6 dB	-4
Duck	lead.duck	0 .. 1	0.2

10. Counter

The counter-lead answers the lead. It plays in the gaps the lead leaves and in a register of its own, so the two never mask each other; Counter Mode on the Set tab (or the style, on Auto) decides whether it echoes, answers or holds against the lead. The engine is the same as the lead's, with its own knobs and its own presets.



The Counter tab.

SOUND

Control	Key	Range	Default
Counter Own Sound	mix.counter_own	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	counter.osc	Supersaw VA FM Wavetable
Table	counter.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo Wav
Position	counter.position	0 .. 1
Detune	counter.detune	0 .. 1
Mix	counter.mix	0 .. 1
Wave	counter.wave	0 .. 1
Dynamic Detune	counter.dynamic_detune	0 .. 1
Pulse Width	counter.pulse_width	0.05 .. 0.95
Drift	counter.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	counter.pos_env	-1 .. 1	0
Position LFO	counter.pos_lfo_depth	0 .. 0.5	0.25
Position Decay	counter.pos_decay	10 .. 8000 ms	1500
LFO Period	counter.pos_lfo_beats	0.25 .. 64 beats	4

FM

Control	Key	Range	Default
FM Ratio	counter.fm_ratio	0.5 .. 8	2
FM Index	counter.fm_index	0 .. 10	2.5
FM Decay	counter.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	counter.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	counter.osc2_mix	0 .. 1	0.3
Osc 2 Interval	counter.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	counter.osc2_detune	-50 .. 50 ct	6

FILTER

Control	Key	Range	Default
Filter Model	counter.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	counter.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	counter.cutoff	200 .. 18000 Hz	5000
Resonance	counter.resonance	0 .. 1	0.2
Env Amount	counter.env_amount	0 .. 6 oct	3.2
Filter Mode	counter.filter_mode	0 .. 1	0
Key Track	counter.key_track	0 .. 1	0.5
HP Floor	counter.hp_floor	40 .. 400 Hz	220
HP Track	counter.hp_track	0 .. 1 x f0	1
Disperse	counter.disperse	0 .. 8 x	0
Disperse Freq	counter.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	counter.filt_attack	0.1 .. 4000 ms	0.1

Filter Decay	counter.filter_decay	5 .. 3000 ms	60
Filter Sustain	counter.filt_sustain	0 .. 1	0
Filter Release	counter.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	counter.amp_attack	0.3 .. 2000 ms	1
Decay	counter.amp_decay	5 .. 4000 ms	70
Sustain	counter.amp_sustain	0 .. 1	0
Release	counter.amp_release	5 .. 4000 ms	35
Velocity	counter.vel_sens	0 .. 1	0.4
Glide	counter.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	counter.menv_attack	0.1 .. 8000 ms	10
Mod Decay	counter.menv_decay	5 .. 12000 ms	800
Mod Sustain	counter.menv_sustain	0 .. 1	0
Mod Release	counter.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	counter.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	counter.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	counter.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	counter.lfo1_retrigger	off / on	off
LFO 1 Fade	counter.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	counter.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	counter.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	counter.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	counter.lfo2_retrigger	off / on	off
LFO 2 Fade	counter.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	counter.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	counter.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	counter.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	counter.lfo3_retrigger	off / on	off
LFO 3 Fade	counter.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	counter.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	counter.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	counter.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	counter.lfo4_retrigger	off / on	off
LFO 4 Fade	counter.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	counter.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	counter.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	counter.mx1_amount	-1 .. 1	0
Mod 2 Source	counter.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off

Mod 2 Target	counter.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	counter.mx2_amount	-1 .. 1	0
Mod 3 Source	counter.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	counter.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	counter.mx3_amount	-1 .. 1	0
Mod 4 Source	counter.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	counter.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	counter.mx4_amount	-1 .. 1	0
Mod 5 Source	counter.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	counter.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	counter.mx5_amount	-1 .. 1	0
Mod 6 Source	counter.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	counter.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	counter.mx6_amount	-1 .. 1	0
Mod 7 Source	counter.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	counter.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	counter.mx7_amount	-1 .. 1	0
Mod 8 Source	counter.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	counter.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	counter.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	counter.lfo_beats	0.25 .. 64 beats	4
LFO to Cutoff	counter.lfo_cutoff	0 .. 4 oct	0.35
LFO to Pitch	counter.lfo_pitch	0 .. 50 ct	0
LFO to Level	counter.lfo_amp	0 .. 1	0

VOICE FX

Control	Key	Range	Default
Modulation	counter.mod	Off Flanger Phaser Comb	Flanger
Mod Mix	counter.mod_mix	0 .. 1	0.28
Mod Depth	counter.mod_depth	0 .. 1	0.75
Mod Period	counter.mod_beats	0.5 .. 64 beats	4
Mod Feedback	counter.mod_feedback	-0.9 .. 0.9	0.55

TRANCE GATE

Control	Key	Range	Default
Trance Gate	counter.gate	off / on	off
Gate Pattern	counter.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	counter.gate_depth	0 .. 1	0.85
Gate Duty	counter.gate_duty	0.05 .. 1	0.5
Gate Attack	counter.gate_attack	0.5 .. 60 ms	3
Gate Release	counter.gate_release	0.5 .. 120 ms	12
Gate Tone	counter.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	counter.delay_send	0 .. 1	0.4
Delay Feedback	counter.delay_feedback	0 .. 0.9	0.4
Delay Left	counter.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	1/16
Delay Right	counter.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/8
Delay High Pass	counter.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	counter.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	counter.distance	0 .. 1	0.5

Width	counter.width	0 .. 1	0.7
Room Send	counter.room_send	0 .. 1	0
Plate Send	counter.plate_send	0 .. 1	0.2
Hall Send	counter.hall_send	0 .. 1	0
Hall Gate	counter.hall_gate	off / on	off
Pan	counter.pan	-1 .. 1	0.2
Slow Movement	counter.slow_mod	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	counter.level	-36 .. 6 dB	-3
Duck	counter.duck	0 .. 1	0.2

11. Arp

The same engine as the lead, with its own knobs. The arpeggio runs on chord tones, and when its pitch range would overlap the lead's by more than two semitones it is octaved away: two voices playing in the same two octaves mask each other, and the composer checks that before it writes the notes.

The counter never strikes together with the lead a semitone or a minor ninth apart on beats 1 and 3.

OSCILLATOR

Control	Key	Range
Oscillator	arp.osc	Supersaw VA FM Wavetable
Table	arp.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo WaveEdit
Position	arp.position	0 .. 1
Detune	arp.detune	0 .. 1
Mix	arp.mix	0 .. 1
Wave	arp.wave	0 .. 1
Dynamic Detune	arp.dynamic_detune	0 .. 1
Pulse Width	arp.pulse_width	0,05 .. 0,95
Drift	arp.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	arp.pos_env	-1 .. 1	0
Position LFO	arp.pos_lfo_depth	0 .. 0,5	0,15
Position Decay	arp.pos_decay	10 .. 8000 ms	1500
LFO Period	arp.pos_lfo_beats	0,25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	arp.fm_ratio	0,5 .. 8	2
FM Index	arp.fm_index	0 .. 10	2,5
FM Decay	arp.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	arp.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	arp.osc2_mix	0 .. 1	0,26
Osc 2 Interval	arp.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	arp.osc2_detune	-50 .. 50 ct	4

FILTER

Control	Key	Range	Default
Filter Model	arp.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	arp.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	arp.cutoff	200 .. 18000 Hz	3500
Resonance	arp.resonance	0 .. 1	0,3
Env Amount	arp.env_amount	0 .. 6 oct	2,8
Filter Mode	arp.filter_mode	0 .. 1	0
Key Track	arp.key_track	0 .. 1	0,5
HP Floor	arp.hp_floor	40 .. 400 Hz	220
HP Track	arp.hp_track	0 .. 1 x f0	1
Disperse	arp.disperse	0 .. 8 x	0
Disperse Freq	arp.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	arp.filt_attack	0,1 .. 4000 ms	0,1
Filter Decay	arp.filter_decay	5 .. 3000 ms	140
Filter Sustain	arp.filt_sustain	0 .. 1	0
Filter Release	arp.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
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Attack	arp.amp_attack	0.3 .. 2000 ms	0.8
Decay	arp.amp_decay	5 .. 4000 ms	220
Sustain	arp.amp_sustain	0 .. 1	0
Release	arp.amp_release	5 .. 4000 ms	12
Velocity	arp.vel_sens	0 .. 1	0.4
Glide	arp.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	arp.menv_attack	0.1 .. 8000 ms	10
Mod Decay	arp.menv_decay	5 .. 12000 ms	800
Mod Sustain	arp.menv_sustain	0 .. 1	0
Mod Release	arp.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	arp.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	arp.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	arp.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	arp.lfo1_retrigger	off / on	off
LFO 1 Fade	arp.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	arp.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	arp.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	arp.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	arp.lfo2_retrigger	off / on	off
LFO 2 Fade	arp.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	arp.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	arp.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	arp.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	arp.lfo3_retrigger	off / on	off
LFO 3 Fade	arp.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	arp.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	arp.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	arp.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	arp.lfo4_retrigger	off / on	off
LFO 4 Fade	arp.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	arp.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	arp.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	arp.mx1_amount	-1 .. 1	0
Mod 2 Source	arp.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	arp.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	arp.mx2_amount	-1 .. 1	0
Mod 3 Source	arp.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	arp.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	arp.mx3_amount	-1 .. 1	0
Mod 4 Source	arp.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off

Mod 4 Target	arp.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	arp.mx4_amount	-1 .. 1	0
Mod 5 Source	arp.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	arp.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	arp.mx5_amount	-1 .. 1	0
Mod 6 Source	arp.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	arp.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	arp.mx6_amount	-1 .. 1	0
Mod 7 Source	arp.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	arp.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	arp.mx7_amount	-1 .. 1	0
Mod 8 Source	arp.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	arp.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	arp.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	arp.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	arp.lfo_cutoff	0 .. 4 oct	0.28
LFO to Pitch	arp.lfo_pitch	0 .. 50 ct	0
LFO to Level	arp.lfo_amp	0 .. 1	0.08

VOICE FX

Control	Key	Range	Default
Modulation	arp.mod	Off Flanger Phaser Comb	Comb
Mod Mix	arp.mod_mix	0 .. 1	0.2
Mod Depth	arp.mod_depth	0 .. 1	0
Mod Period	arp.mod_beats	0.5 .. 64 beats	16
Mod Feedback	arp.mod_feedback	-0.9 .. 0.9	0.6

TRANCE GATE

Control	Key	Range	Default
Trance Gate	arp.gate	off / on	off
Gate Pattern	arp.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	arp.gate_depth	0 .. 1	0.85
Gate Duty	arp.gate_duty	0.05 .. 1	0.5
Gate Attack	arp.gate_attack	0.5 .. 60 ms	3
Gate Release	arp.gate_release	0.5 .. 120 ms	12
Gate Tone	arp.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	arp.delay_send	0 .. 1	0.35
Delay Feedback	arp.delay_feedback	0 .. 0.9	0.4
Delay Left	arp.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	arp.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/8
Delay High Pass	arp.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	arp.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	arp.distance	0 .. 1	0.5
Width	arp.width	0 .. 1	0.6
Room Send	arp.room_send	0 .. 1	0
Plate Send	arp.plate_send	0 .. 1	0.2
Hall Send	arp.hall_send	0 .. 1	0
Hall Gate	arp.hall_gate	off / on	off
Pan	arp.pan	-1 .. 1	0

Slow Movement	arp.slow_mod	0 .. 1	0
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OUTPUT

Control	Key	Range	Default
Level	arp.level	-36 .. 6 dB	-5
Duck	arp.duck	0 .. 1	0.2

12. Stab

Short chord hits on the off-beats and the phrase ends, the punctuation of a groove. They are voiced from the track's harmony and kept out of the lead's register.

PHOSPHENE Progressive F# Harmonic Minor Set 60 min Compose set New seed Stop Mute

seed 112484790 145.0 BPM own clock playing track 1, bar 16 (bar 16 of the set)

ARRANGE 24 tracks 5456 bars planned set 60 min

Set Arrange Low End Percussion Acid **Synths** Effects Mixer Perform Export Gallery

Lead Counter **Arp** **Stab** Pad Drone

SOUND

Stab Own Sound Save...
Supersaw Stab: Buried Torch (composer)
Preset

OSCILLATOR

Supersaw Vocal
Oscillator Table
Detune 0.74 Mix 0.83 Wave 0.00 Dynamic Detune 0.00 Position 0.30
Drift 4.87 Hz Pulse Width 0.50

WAVETABLE MOTION

Position Env 0.00 Position LFO 0.15
Position Decay 0.00 LFO Period 0.00

FM

FM Ratio 2.00 FM Index 2.50
FM Decay 200 ms

SECOND OSCILLATOR

Off Oscillator 2
-1 Oct Osc 2 Detune 0.26 Osc 2 Mix 0.00
Osc 2 Interval 0.00

MOD ENVELOPE

Mod Attack 0.00 ms Mod Decay 0.00 ms Mod Sustain 0.00 ms Mod Release 0.00 ms

FILTER

Prophet (SSM2040) Low Pass
Filter Model Filter Type
Resonance 0.13 Env Amount 1.93 oct Filter Mode 0.00
HP Floor 0.00 Key Track 0.50 Cutoff 2500 Hz
Disperse Freq 0.00 HP Track 0.00 Disperse 0.00

FILTER ENVELOPE

Filter Attack 0.00 ms Filter Decay 0.00 ms Filter Sustain 0.00 ms Filter Release 0.00 ms

AMPLIFIER

Attack 2.84 ms Decay 335.284 ms Sustain 0.83 Release 122.574 ms
Velocity 0.40 Glide 0.13 ms

LFO 2

Sine Free
LFO 2 Shape LFO 2 Sync 1.00 Hz
LFO 2 Retrigger 0.00 s LFO 2 Fade 0.00 s

LFO 3

Sine Free
LFO 3 Shape LFO 3 Sync 1.00 Hz
LFO 3 Retrigger 0.00 s LFO 3 Fade 0.00 s

LFO 1

Sine Free
LFO 1 Shape LFO 1 Sync 0.13 Hz
LFO 1 Retrigger 0.00 s LFO 1 Fade 0.00 s

LFO 4

Sine Free
LFO 4 Shape LFO 4 Sync 1.00 Hz
LFO 4 Retrigger 0.00 s LFO 4 Fade 0.00 s

MOD MATRIX

LFO 1	Cutoff	Velocity	Cutoff		
Mod 1 Source	Mod 1 Target	Mod 1 Amount	Mod 2 Source	Mod 2 Target	Mod 2 Amount
Off	Off	0.00	Off	Off	0.00
Mod 3 Source	Mod 3 Target	Mod 3 Amount	Mod 4 Source	Mod 4 Target	Mod 4 Amount
Off	Off	0.00	Off	Off	0.00
Mod 5 Source	Mod 5 Target	Mod 5 Amount	Mod 6 Source	Mod 6 Target	Mod 6 Amount
Off	Off	0.00	Off	Off	0.00
Mod 7 Source	Mod 7 Target	Mod 7 Amount	Mod 8 Source	Mod 8 Target	Mod 8 Amount
Off	Off	0.00	Off	Off	0.00

TRANCE GATE

Trance Gate Gate Pattern Sixteenths Gate Depth 0.85
Gate Duty 0.50 Gate Attack 0.00 ms Gate Release 0.00 ms Gate Tone 0.40

DELAY

Delay Send 0.45 Delay Feedback 0.45
Delay Left 3/16 Delay High Pass 300 Hz
Delay Right 1/4 Delay Low Pass 600 Hz

SPACE & IMAGE

Distance 0.50 Width 0.90
Room Send 0.00 Plate Send 0.15 Hall Send 0.00
Hall Gate 0.00 Pan 0.00 Slow Movement 0.00

VOICE FX

Off Modulation 0.00 Mod Mix 0.70 Mod Depth 0.00
Mod Feedback 0.50

OUTPUT

Level -7.00 dB Duck 0.20

PATTERN

STAB bars 17 to 20

this part is silent here

The Stab tab.

SOUND

Control	Key	Range	Default
Stab Own Sound	mix.stab_own	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	stab.osc	Supersaw VA FM Wavetable
Table	stab.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo WaveEd
Position	stab.position	0 .. 1
Detune	stab.detune	0 .. 1
Mix	stab.mix	0 .. 1
Wave	stab.wave	0 .. 1
Dynamic Detune	stab.dynamic_detune	0 .. 1
Pulse Width	stab.pulse_width	0.05 .. 0.95
Drift	stab.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	stab.pos_env	-1 .. 1	0
Position LFO	stab.pos_lfo_depth	0 .. 0.5	0.15
Position Decay	stab.pos_decay	10 .. 8000 ms	1500
LFO Period	stab.pos_lfo_beats	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	stab.fm_ratio	0.5 .. 8	2
FM Index	stab.fm_index	0 .. 10	2.5
FM Decay	stab.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	stab.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	stab.osc2_mix	0 .. 1	0.26
Osc 2 Interval	stab.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	stab.osc2_detune	-50 .. 50 ct	5

FILTER

Control	Key	Range	Default
Filter Model	stab.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	stab.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	stab.cutoff	200 .. 18000 Hz	1800
Resonance	stab.resonance	0 .. 1	0.35
Env Amount	stab.env_amount	0 .. 6 oct	3.5
Filter Mode	stab.filter_mode	0 .. 1	0
Key Track	stab.key_track	0 .. 1	0.5
HP Floor	stab.hp_floor	40 .. 400 Hz	220
HP Track	stab.hp_track	0 .. 1 x f0	1
Disperse	stab.disperse	0 .. 8 x	0
Disperse Freq	stab.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	stab.filt_attack	0.1 .. 4000 ms	0.1
Filter Decay	stab.filter_decay	5 .. 3000 ms	90
Filter Sustain	stab.filt_sustain	0 .. 1	0

Filter Release	stab.filt_release	5 .. 8000 ms	300
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AMPLIFIER

Control	Key	Range	Default
Attack	stab.amp_attack	0.3 .. 2000 ms	0.5
Decay	stab.amp_decay	5 .. 4000 ms	180
Sustain	stab.amp_sustain	0 .. 1	0
Release	stab.amp_release	5 .. 4000 ms	60
Velocity	stab.vel_sens	0 .. 1	0.4
Glide	stab.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	stab.menv_attack	0.1 .. 8000 ms	10
Mod Decay	stab.menv_decay	5 .. 12000 ms	800
Mod Sustain	stab.menv_sustain	0 .. 1	0
Mod Release	stab.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	stab.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	stab.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	stab.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	stab.lfo1_retrigger	off / on	off
LFO 1 Fade	stab.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	stab.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	stab.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	stab.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	stab.lfo2_retrigger	off / on	off
LFO 2 Fade	stab.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	stab.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	stab.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	stab.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	stab.lfo3_retrigger	off / on	off
LFO 3 Fade	stab.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	stab.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	stab.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	stab.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	stab.lfo4_retrigger	off / on	off
LFO 4 Fade	stab.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	stab.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	stab.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	stab.mx1_amount	-1 .. 1	0
Mod 2 Source	stab.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	stab.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	stab.mx2_amount	-1 .. 1	0

Mod 3 Source	stab.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	stab.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	stab.mx3_amount	-1 .. 1	0
Mod 4 Source	stab.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	stab.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	stab.mx4_amount	-1 .. 1	0
Mod 5 Source	stab.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	stab.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	stab.mx5_amount	-1 .. 1	0
Mod 6 Source	stab.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	stab.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	stab.mx6_amount	-1 .. 1	0
Mod 7 Source	stab.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	stab.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	stab.mx7_amount	-1 .. 1	0
Mod 8 Source	stab.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	stab.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	stab.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	stab.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	stab.lfo_cutoff	0 .. 4 oct	0.22
LFO to Pitch	stab.lfo_pitch	0 .. 50 ct	0
LFO to Level	stab.lfo_amp	0 .. 1	0

VOICE FX

Control	Key	Range	Default
Modulation	stab.mod	Off Flanger Phaser Comb	Off
Mod Mix	stab.mod_mix	0 .. 1	0
Mod Depth	stab.mod_depth	0 .. 1	0.7
Mod Period	stab.mod_beats	0.5 .. 64 beats	16
Mod Feedback	stab.mod_feedback	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	stab.gate	off / on	off
Gate Pattern	stab.gate_pattern	Sixteenths Eighthths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	stab.gate_depth	0 .. 1	0.85
Gate Duty	stab.gate_duty	0.05 .. 1	0.5
Gate Attack	stab.gate_attack	0.5 .. 60 ms	3
Gate Release	stab.gate_release	0.5 .. 120 ms	12
Gate Tone	stab.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	stab.delay_send	0 .. 1	0.45
Delay Feedback	stab.delay_feedback	0 .. 0.9	0.45
Delay Left	stab.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	stab.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay High Pass	stab.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	stab.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	stab.distance	0 .. 1	0.5
Width	stab.width	0 .. 1	0.9
Room Send	stab.room_send	0 .. 1	0

Plate Send	stab.plate_send	0 .. 1	0.15
Hall Send	stab.hall_send	0 .. 1	0
Hall Gate	stab.hall_gate	off / on	off
Pan	stab.pan	-1 .. 1	0
Slow Movement	stab.slow_mod	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	stab.level	-36 .. 6 dB	-7
Duck	stab.duck	0 .. 1	0.2

13. Pad

Four-voice voicings with the smallest voice movement between chords, through the wavetable engine. The pads carry the blocks that have no lead and no acid, and the trance gate gives them their movement.

Every pitched voice plays its FM ratio to the nearest half (1, 1.5, 2, ...), so an FM pad or lead is always harmonic: a ratio in between makes the bell and struck-metal spectra that sounded out of tune under a chord. A second oscillator stays within a few cents of its interval, and the pad's partner plays in unison or an octave up, never a fifth, which would stack a second chord on the first.

A pad never holds a minor ninth -- no note a semitone above the tonic that bass and drone play, none a semitone above its own root -- nor the leading tone under the tonic, nor a tritone over its root. Held for bars, those intervals sound wrong rather than dark; the genre makes its dissonance with the sound (clusters, noise, drones), and the Phrygian b2 belongs to the lead and the acid. Where the progression swings to the bII, the pad stays on the tonic and adds the minor sixth instead (E - B - E - C over E). The major seventh is a rare colour, on bVI and bIII only.



The Pad tab.

SOUND

Control	Key	Range	Default
Pad Own Sound	mix.pad_own	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	pad.osc	Supersaw VA FM Wavetable
Table	pad.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo WaveEdit
Position	pad.position	0 .. 1
Detune	pad.detune	0 .. 1
Mix	pad.mix	0 .. 1
Wave	pad.wave	0 .. 1
Dynamic Detune	pad.dynamic_detune	0 .. 1
Pulse Width	pad.pulse_width	0,05 .. 0,95
Drift	pad.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	pad.pos_env	-1 .. 1	0.3
Position LFO	pad.pos_lfo_depth	0 .. 0.5	0.15
Position Decay	pad.pos_decay	10 .. 8000 ms	3000
LFO Period	pad.pos_lfo_beats	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	pad.fm_ratio	0.5 .. 8	2
FM Index	pad.fm_index	0 .. 10	2.5
FM Decay	pad.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	pad.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	pad.osc2_mix	0 .. 1	0.38
Osc 2 Interval	pad.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	pad.osc2_detune	-50 .. 50 ct	9

FILTER

Control	Key	Range	Default
Filter Model	pad.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	pad.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	pad.cutoff	200 .. 18000 Hz	5000
Resonance	pad.resonance	0 .. 1	0.1
Env Amount	pad.env_amount	0 .. 6 oct	0
Filter Mode	pad.filter_mode	0 .. 1	0
Key Track	pad.key_track	0 .. 1	0.5
HP Floor	pad.hp_floor	40 .. 400 Hz	140
HP Track	pad.hp_track	0 .. 1 x f0	1
Disperse	pad.disperse	0 .. 8 x	0
Disperse Freq	pad.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	pad.filt_attack	0.1 .. 4000 ms	0.1
Filter Decay	pad.filter_decay	5 .. 3000 ms	400
Filter Sustain	pad.filt_sustain	0 .. 1	0
Filter Release	pad.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
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Attack	pad.amp_attack	0.3 .. 2000 ms	700
Decay	pad.amp_decay	5 .. 4000 ms	2000
Sustain	pad.amp_sustain	0 .. 1	1
Release	pad.amp_release	5 .. 4000 ms	1800
Velocity	pad.vel_sens	0 .. 1	0.4
Glide	pad.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	pad.menv_attack	0.1 .. 8000 ms	10
Mod Decay	pad.menv_decay	5 .. 12000 ms	800
Mod Sustain	pad.menv_sustain	0 .. 1	0
Mod Release	pad.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	pad.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	pad.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	pad.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	pad.lfo1_retrigger	off / on	off
LFO 1 Fade	pad.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	pad.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	pad.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	pad.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	pad.lfo2_retrigger	off / on	off
LFO 2 Fade	pad.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	pad.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	pad.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	pad.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	pad.lfo3_retrigger	off / on	off
LFO 3 Fade	pad.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	pad.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	pad.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	pad.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	pad.lfo4_retrigger	off / on	off
LFO 4 Fade	pad.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	pad.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	pad.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	pad.mx1_amount	-1 .. 1	0
Mod 2 Source	pad.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	pad.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	pad.mx2_amount	-1 .. 1	0
Mod 3 Source	pad.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	pad.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	pad.mx3_amount	-1 .. 1	0
Mod 4 Source	pad.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off

Mod 4 Target	pad.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	pad.mx4_amount	-1 .. 1	0
Mod 5 Source	pad.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	pad.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	pad.mx5_amount	-1 .. 1	0
Mod 6 Source	pad.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	pad.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	pad.mx6_amount	-1 .. 1	0
Mod 7 Source	pad.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	pad.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	pad.mx7_amount	-1 .. 1	0
Mod 8 Source	pad.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	pad.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	pad.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	pad.lfo_beats	0.25 .. 64 beats	16
LFO to Cutoff	pad.lfo_cutoff	0 .. 4 oct	0.5
LFO to Pitch	pad.lfo_pitch	0 .. 50 ct	0
LFO to Level	pad.lfo_amp	0 .. 1	0.1

VOICE FX

Control	Key	Range	Default
Modulation	pad.mod	Off Flanger Phaser Comb	Off
Mod Mix	pad.mod_mix	0 .. 1	0
Mod Depth	pad.mod_depth	0 .. 1	0.7
Mod Period	pad.mod_beats	0.5 .. 64 beats	16
Mod Feedback	pad.mod_feedback	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	pad.gate	off / on	off
Gate Pattern	pad.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	pad.gate_depth	0 .. 1	0.85
Gate Duty	pad.gate_duty	0.05 .. 1	0.5
Gate Attack	pad.gate_attack	0.5 .. 60 ms	3
Gate Release	pad.gate_release	0.5 .. 120 ms	12
Gate Tone	pad.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	pad.delay_send	0 .. 1	0
Delay Feedback	pad.delay_feedback	0 .. 0.9	0.4
Delay Left	pad.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	pad.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay High Pass	pad.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	pad.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	pad.distance	0 .. 1	1
Width	pad.width	0 .. 1	1
Room Send	pad.room_send	0 .. 1	0
Plate Send	pad.plate_send	0 .. 1	0.35
Hall Send	pad.hall_send	0 .. 1	0
Hall Gate	pad.hall_gate	off / on	off
Pan	pad.pan	-1 .. 1	0

Slow Movement	<code>pad.slow_mod</code>	0 .. 1	0.5
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OUTPUT

Control	Key	Range	Default
Level	<code>pad.level</code>	-36 .. 6 dB	-16
Duck	<code>pad.duck</code>	0 .. 1	0.5

14. Drone

A long, slowly moving tone under the breakdowns and the intros: the sound that holds a track together when the kick is gone. It is high-passed above the bass so it never takes the floor. It holds the tonic, its fifth and its octave, and like the pad never a minor ninth over the bass -- in a DJ blend a note of the incoming track that would stand a semitone over the outgoing tonic waits for the hand-over. With Bass Follows Chords on it follows the bass to the pad's root.



The Drone tab.

SOUND

Control	Key	Range	Default
Drone Own Sound	mix.drone_own	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	drone.osc	Supersaw VA FM Wavetable
Table	drone.table	Classic Vocal Glass PWM Sync Formant Saw WaveEdit Hienharm WaveEdit Junox_ho WaveEdit Euclidea WaveEdit Sohler49 WaveEdit Vocal_fo WaveE
Position	drone.position	0 .. 1
Detune	drone.detune	0 .. 1
Mix	drone.mix	0 .. 1
Wave	drone.wave	0 .. 1
Dynamic Detune	drone.dynamic_detune	0 .. 1
Pulse Width	drone.pulse_width	0.05 .. 0.95
Drift	drone.drift	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	drone.pos_env	-1 .. 1	0
Position LFO	drone.pos_lfo_depth	0 .. 0.5	0.1
Position Decay	drone.pos_decay	10 .. 8000 ms	1500
LFO Period	drone.pos_lfo_beats	0.25 .. 64 beats	32

FM

Control	Key	Range	Default
FM Ratio	drone.fm_ratio	0.5 .. 8	2
FM Index	drone.fm_index	0 .. 10	2.5
FM Decay	drone.fm_decay	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	drone.osc2	Off Supersaw VA FM Wavetable	Off
Osc 2 Mix	drone.osc2_mix	0 .. 1	0.36
Osc 2 Interval	drone.osc2_interval	-2 Oct -1 Oct -5th Unison +5th +1 Oct	-1 Oct
Osc 2 Detune	drone.osc2_detune	-50 .. 50 ct	5

FILTER

Control	Key	Range	Default
Filter Model	drone.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	drone.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	drone.cutoff	200 .. 18000 Hz	900
Resonance	drone.resonance	0 .. 1	0.1
Env Amount	drone.env_amount	0 .. 6 oct	0
Filter Mode	drone.filter_mode	0 .. 1	0
Key Track	drone.key_track	0 .. 1	0.5
HP Floor	drone.hp_floor	40 .. 400 Hz	140
HP Track	drone.hp_track	0 .. 1 x f0	0.5
Disperse	drone.disperse	0 .. 8 x	0
Disperse Freq	drone.disperse_freq	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	drone.filt_attack	0.1 .. 4000 ms	0.1

Filter Decay	drone.filter_decay	5 .. 3000 ms	400
Filter Sustain	drone.filt_sustain	0 .. 1	0
Filter Release	drone.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	drone.amp_attack	0.3 .. 2000 ms	1500
Decay	drone.amp_decay	5 .. 4000 ms	2000
Sustain	drone.amp_sustain	0 .. 1	1
Release	drone.amp_release	5 .. 4000 ms	2500
Velocity	drone.vel_sens	0 .. 1	0.4
Glide	drone.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	drone.menv_attack	0.1 .. 8000 ms	10
Mod Decay	drone.menv_decay	5 .. 12000 ms	800
Mod Sustain	drone.menv_sustain	0 .. 1	0
Mod Release	drone.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	drone.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	drone.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	drone.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	drone.lfo1_retrigger	off / on	off
LFO 1 Fade	drone.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	drone.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	drone.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	drone.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	drone.lfo2_retrigger	off / on	off
LFO 2 Fade	drone.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	drone.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	drone.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	drone.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	drone.lfo3_retrigger	off / on	off
LFO 3 Fade	drone.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	drone.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	drone.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	drone.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	drone.lfo4_retrigger	off / on	off
LFO 4 Fade	drone.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	drone.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	drone.mx1_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 1 Amount	drone.mx1_amount	-1 .. 1	0
Mod 2 Source	drone.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off

Mod 2 Target	drone.mx2_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 2 Amount	drone.mx2_amount	-1 .. 1	0
Mod 3 Source	drone.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	drone.mx3_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 3 Amount	drone.mx3_amount	-1 .. 1	0
Mod 4 Source	drone.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	drone.mx4_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 4 Amount	drone.mx4_amount	-1 .. 1	0
Mod 5 Source	drone.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	drone.mx5_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 5 Amount	drone.mx5_amount	-1 .. 1	0
Mod 6 Source	drone.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	drone.mx6_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 6 Amount	drone.mx6_amount	-1 .. 1	0
Mod 7 Source	drone.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	drone.mx7_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 7 Amount	drone.mx7_amount	-1 .. 1	0
Mod 8 Source	drone.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	drone.mx8_dst	Off Pitch Osc 2 Pitch Pulse Width Table Position FM Index Cutoff Resonance Filter Mode Level Pan	Off
Mod 8 Amount	drone.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	drone.lfo_beats	0.25 .. 64 beats	32
LFO to Cutoff	drone.lfo_cutoff	0 .. 4 oct	0.32
LFO to Pitch	drone.lfo_pitch	0 .. 50 ct	0
LFO to Level	drone.lfo_amp	0 .. 1	0.08

VOICE FX

Control	Key	Range	Default
Modulation	drone.mod	Off Flanger Phaser Comb	Off
Mod Mix	drone.mod_mix	0 .. 1	0
Mod Depth	drone.mod_depth	0 .. 1	0.7
Mod Period	drone.mod_beats	0.5 .. 64 beats	16
Mod Feedback	drone.mod_feedback	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	drone.gate	off / on	off
Gate Pattern	drone.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	drone.gate_depth	0 .. 1	0.85
Gate Duty	drone.gate_duty	0.05 .. 1	0.5
Gate Attack	drone.gate_attack	0.5 .. 60 ms	3
Gate Release	drone.gate_release	0.5 .. 120 ms	12
Gate Tone	drone.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	drone.delay_send	0 .. 1	0
Delay Feedback	drone.delay_feedback	0 .. 0.9	0.4
Delay Left	drone.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	drone.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay High Pass	drone.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	drone.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	drone.distance	0 .. 1	1

Width	drone.width	0 .. 1	1
Room Send	drone.room_send	0 .. 1	0
Plate Send	drone.plate_send	0 .. 1	0.5
Hall Send	drone.hall_send	0 .. 1	0
Hall Gate	drone.hall_gate	off / on	off
Pan	drone.pan	-1 .. 1	0
Slow Movement	drone.slow_mod	0 .. 1	0.25

OUTPUT

Control	Key	Range	Default
Level	drone.level	-36 .. 6 dB	-17
Duck	drone.duck	0 .. 1	0.45

15. SFX / FX

The effect generator makes risers, downlifters, impacts, sweeps, formant shots, reverse swells and zaps; the composer places them at the form's boundaries -- a riser over the last eight bars of a buildup, ending exactly on the drop, the formant shot on the last beat of the pre-drop break, a descending sweep into the drop.

Short one-shots -- zaps, squelches, bubbles, alien chatter -- fall at the ends of phrases, about one in eight bars of a groove and one in four of the climax; a groove is never left eight bars without an effect, and what fills such a gap is a sweep or a swell, never a zap.

Effect Presets: every family of the effect bank (2048 presets in eleven families: riser, downlifter, impact, sweep, formant shot, reverse swell, zap, squelch, bubble, reverse crash, atmosphere) has a chooser. Auto lets the generator draw a preset of the family for every event, never the same twice in a track; a number makes every event of that family play that preset. "Hear" plays the chosen preset in the running set (with Auto, one of the family at random); while the transport stands nothing sounds.

Space Shot: a recording of NASA's, up to 25 seconds -- a rocket launch, Sputnik's beep, Saturn's radio emissions, a quake on Mars -- played whole, once, on this strip: after the cut of a breakdown, now and then in a long intro, most in Full-On and Hi-Tech, and never within two bars of a spoken phrase. It comes from the field recordings (Field tab); without them it stays silent.

The tonal layer of an effect -- an atmosphere's held tone above all -- is mapped into the mode, and never onto the b2 or the leading tone: a minor seventh the mode lacks becomes the fifth, not the major seventh.

The two send effects are a short room and a long hall, both feedback delay networks. Their returns are ducked with the kick like everything else, and high-passed: a reverb tail in the bass register is mud.



The SFX / FX tab.

EFFECT GENERATOR

Control	Key	Range	Default
Level	sfx.level	-36 .. 6 dB	-3
Noise	sfx.noise	0 .. 1	0.6
Resonance	sfx.resonance	0 .. 1	0.5
Brightness	sfx.brightness	0 .. 1	0.5
Impact Decay	sfx.impact_decay	200 .. 4000 ms	1400
Swell Decay	sfx.swell_decay	200 .. 6000 ms	1500
Vowel	sfx.vowel	0 .. 1	0.3
Width	sfx.width	0 .. 1	0.7

Sub Drop	sfx.sub_level	-36 .. 6 dB	-14
Sub Duck	sfx.sub_duck	0 .. 1	1
Wander	sfx.wander	off / on	off
Wander Send	sfx.wander_send	0 .. 1	0.85
Room Send	sfx.room_send	0 .. 1	0
Plate Send	sfx.plate_send	0 .. 1	0.1
Hall Send	sfx.hall_send	0 .. 1	0.5
Duck	sfx.duck	0 .. 1	0

EFFECT PRESETS

Control	Key	Range	Default
Riser	sfx.preset_riser	0 .. 256	0
Downlifter	sfx.preset_downlifter	0 .. 128	0
Impact	sfx.preset_impact	0 .. 128	0
Sweep	sfx.preset_sweep	0 .. 256	0
Formant Shot	sfx.preset_formant_shot	0 .. 96	0
Reverse Swell	sfx.preset_reverse_swell	0 .. 256	0
Zap	sfx.preset_zap	0 .. 96	0
Squelch	sfx.preset_squelch	0 .. 128	0
Bubble	sfx.preset_bubble	0 .. 64	0
Reverse Crash	sfx.preset_reverse_crash	0 .. 128	0
Atmosphere	sfx.preset_atmosphere	0 .. 512	0

ROOM (A)

Control	Key	Range	Default
Room Size	fx.room_size	0.3 .. 3	0.5
Room Decay	fx.room_decay	0.1 .. 4 s	0.7
Room Damping	fx.room_damping	0 .. 1	0.5
Room Pre-Delay	fx.room_pre_delay	0 .. 100 ms	50
Room Low Cut	fx.room_low_cut	100 .. 1000 Hz	200
Room High Cut	fx.room_high_cut	1000 .. 20000 Hz	10000

PLATE (B)

Control	Key	Range	Default
Plate Size	fx.plate_size	0.3 .. 3	0.8
Plate Decay	fx.plate_decay	0.2 .. 8 s	1.5
Plate Damping	fx.plate_damping	0 .. 1	0.35
Plate Pre-Delay	fx.plate_pre_delay	0 .. 100 ms	25
Plate Low Cut	fx.plate_low_cut	100 .. 1000 Hz	300
Plate High Cut	fx.plate_high_cut	1000 .. 20000 Hz	6000
Plate to Hall	fx.plate_to_hall	0 .. 1	0.15

HALL (C)

Control	Key	Range	Default
Hall Size	fx.hall_size	0.3 .. 3	1.6
Hall Decay	fx.hall_decay	0.3 .. 20 s	10
Hall Damping	fx.hall_damping	0 .. 1	0.45
Hall Pre-Delay	fx.hall_pre_delay	0 .. 0.5 beats	0.0125
Hall Low Cut	fx.low_cut	150 .. 1000 Hz	400
Hall High Cut	fx.high_cut	1000 .. 20000 Hz	3000

RETURNS

Control	Key	Range	Default
Room Return	fx.room_return	-36 .. 6 dB	-6
Plate Return	fx.plate_return	-36 .. 6 dB	-6
Hall Return	fx.hall_return	-36 .. 6 dB	-6
Return Duck	fx.return_duck	0 .. 1	0.45
Return Duck Release	fx.return_duck_release	10 .. 800 ms	100

PSY FX

Control	Key	Range	Default
Flanger Period	psyfx.flanger_beats	0.5 .. 32 beats	8
Flanger Depth	psyfx.flanger_depth	0 .. 1	0.7
Flanger Feedback	psyfx.flanger_feedback	-0.9 .. 0.9	0.6
Flanger Mix	psyfx.flanger_mix	0 .. 1	0.35
Phaser Period	psyfx.phaser_beats	0.5 .. 32 beats	16
Phaser Depth	psyfx.phaser_depth	0 .. 1	0.8
Phaser Feedback	psyfx.phaser_feedback	0 .. 0.9	0.5
Phaser Mix	psyfx.phaser_mix	0 .. 1	0.3
Frequency Shift	psyfx.shift_hz	-500 .. 500 Hz	0
Shifter Mix	psyfx.shift_mix	0 .. 1	0.5
Send Return	psyfx.return	-36 .. 6 dB	-3
Motion	psyfx.motion	0 .. 1	0.7

SHAMANIC BED

Control	Key	Range	Default
Width	texture.width	0 .. 1	0.8
Bowl Decay	texture.bowl_decay	1 .. 20 s	7
Bowl Brightness	texture.bowl_bright	0 .. 1	0.4
Didge Formant	texture.didge_formant	0 .. 1	0.5
Didge Breath	texture.didge_breath	0 .. 1	0.5
Jaw Harp Sweep	texture.jaw_sweep	0 .. 1	0.6
Room Send	texture.room_send	0 .. 1	0.1
Hall Send	texture.hall_send	0 .. 1	0.35
FX Send	texture.fx_send	0 .. 1	0.3
Duck	texture.duck	0 .. 1	0.45

VOICES

Control	Key	Range	Default
Pitch Spread	vocal.pitch	0 .. 12 st	3
Drive	vocal.drive	0 .. 1	0.35
Delay Throw	vocal.throw_send	0 .. 1	0.6
Throw Time	vocal.throw_beats	1/16 1/8 3/16 1/4 3/8 1/2	3/8
Throw Feedback	vocal.throw_feedback	0 .. 0.9	0.55
FX Send	vocal.fx_send	0 .. 1	0.45
Hall Send	vocal.hall_send	0 .. 1	0.3
Duck	vocal.duck	0 .. 1	0.4
Width	vocal.width	0 .. 1	0.5

16. Field

The Field track plays field recordings: a place under the music -- a rainforest, a night full of insects, rain on a roof, a river, a cave, a machine hall, the wind on Mars. It is a sampler, and its recordings are the original files, not converted: 120 recordings in 30 categories (four each, the steadiest and most different of their category; generated with Stable Audio 3 for Noctuary) and NASA's sounds from space. They are an optional part of the installation (2 GB). Without them everything plays; only this track and the NASA shots are silent, and the Library line says so.

Where it plays: under the intro from its first bar (so the place changes with the track, over the DJ's blend), over a breakdown, in the outro up to the blend, and -- in Dark Forest, rarely in Progressive -- sixteen quiet bars under a long groove. Never in a build-up or a drop. How often follows the style: Dark Forest always has its place, Hi-Tech least. `compose.field_density` scales it.

What plays: with A Category on Auto the composer draws a preset for every track by its style -- forest, night, insects and swamps for Dark Forest, sea, fire and ritual for Goa, wind, rain and water for Progressive, machines and radio for Hi-Tech, space for Full-On -- and sends it as it does the synths' presets: the knobs stay as they are. Choose a category and the knobs play, in every track.

Preset: loads one of the factory presets into the knobs -- one per recording (by category) and the scenes, two recordings as one place (Jungle Night: a night and its insects; Swamp: the wetland and its mud). The category is then no longer Auto; set it back to hand the choice to the composer again. Library: how many recordings were found, and where. Folder...: a folder of your own recordings beside the installed ones -- FLAC files named `fr-<category>-<anything>.flac` (Noctuary's whole library of 2938 fits as it is); cancel the dialog to remove it.

Layer A and B: each a recording, chosen by category and variation (the variations of a category are its files in name order), with its level, its pitch (semitones and cents, by resampling), where it starts and how far that start may wander per event, and its direction. B is off by default; Layer Mix weighs the two (the middle plays both whole).

Loop: over any stretch of the recording (Loop Start to Loop End), joined by a crossfade of its own length -- over the last stretch before the loop's end a second read head runs from the loop's start, and the two are mixed with equal power, because two stretches of a field recording are uncorrelated and keep their loudness only so. What is heard at the seam is always a blend, never a cut, so a recording that was not made as a loop plays as one. Without the loop a layer ends with its recording.

Filter, Filter Envelope, Amplifier and the modulation block are the synths' (Filters and modulation); the matrix reaches pitch, cutoff, resonance, the filter's mode, level and pan. The Low Cut (24 dB per octave) keeps the recordings out of the range that belongs to kick and bass -- wind and water have most of their power there. The long amp envelope (2 s in, 4 s out) is how a place enters and leaves.

The recordings are loaded when a track is planned, a track ahead of their first note, and kept in memory; an offline render waits for them, so it is the same every time.



The Field tab.

LAYER A

Control	Key	Range
A Category	field.a_category	Auto Rainforest Night Country Insects Foliage Wetland Mud Bubbles Cave River Sea Rain on Land Rain on Roof Wind Underwater Ice and Snow
A Variation	field.a_variation	0 .. 63
A Level	field.a_level	-36 .. 6 dB
A Pitch	field.a_pitch	-24 .. 24 st
A Fine	field.a_fine	-100 .. 100 ct
A Start	field.a_start	0 .. 1
A Start Random	field.a_start_random	0 .. 1

A Reverse	field.a_reverse	off / on
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LAYER B

Control	Key	Range
B On	field.b_on	off / on
B Category	field.b_category	Auto Rainforest Night Country Insects Foliage Wetland Mud Bubbles Cave River Sea Rain on Land Rain on Roof Wind Underwater Ice and Snow
B Variation	field.b_variation	0 .. 63
B Level	field.b_level	-36 .. 6 dB
B Pitch	field.b_pitch	-24 .. 24 st
B Fine	field.b_fine	-100 .. 100 ct
B Start	field.b_start	0 .. 1
B Start Random	field.b_start_random	0 .. 1
B Reverse	field.b_reverse	off / on
Layer Mix	field.layer_mix	0 .. 1

LOOP

Control	Key	Range	Default
Loop	field.loop	off / on	on
Loop Start	field.loop_start	0 .. 1	0
Loop End	field.loop_end	0 .. 1	1
Loop Crossfade	field.loop_xfade	10 .. 10000 ms	1500

FILTER

Control	Key	Range	Default
Filter Model	field.filter_model	State Variable Moog Ladder Prophet (SSM2040) Juno (IR3109) Oberheim SEM Xpander Diode Ladder Korg35 (MS-20) Polivoks Wasp	State Variable
Filter Type	field.filter_type	Low Pass Band Pass High Pass Notch	Low Pass
Cutoff	field.cutoff	40 .. 20000 Hz	20000
Resonance	field.resonance	0 .. 1	0
Env Amount	field.env_amount	-8 .. 8 oct	0
Filter Mode	field.filter_mode	0 .. 1	0
Low Cut	field.low_cut	20 .. 1000 Hz	150

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	field.filt_attack	0.1 .. 4000 ms	0.1
Filter Decay	field.filter_decay	5 .. 30000 ms	2000
Filter Sustain	field.filt_sustain	0 .. 1	0
Filter Release	field.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	field.amp_attack	1 .. 30000 ms	2000
Decay	field.amp_decay	5 .. 30000 ms	1000
Sustain	field.amp_sustain	0 .. 1	1
Release	field.amp_release	5 .. 30000 ms	4000

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	field.menv_attack	0.1 .. 8000 ms	10
Mod Decay	field.menv_decay	5 .. 12000 ms	800
Mod Sustain	field.menv_sustain	0 .. 1	0
Mod Release	field.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	field.lfo1_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 1 Sync	field.lfo1_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 1 Rate	field.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrigger	field.lfo1_retrigger	off / on	off
LFO 1 Fade	field.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	field.lfo2_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 2 Sync	field.lfo2_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 2 Rate	field.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrigger	field.lfo2_retrigger	off / on	off
LFO 2 Fade	field.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	field.lfo3_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 3 Sync	field.lfo3_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 3 Rate	field.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrigger	field.lfo3_retrigger	off / on	off
LFO 3 Fade	field.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	field.lfo4_shape	Sine Triangle Saw Up Saw Down Square Sample & Hold Smooth Random	Sine
LFO 4 Sync	field.lfo4_sync	Free 4 Bars 2 Bars 1 Bar 1/2 1/4 1/8 1/16 1/4 T 1/8 T	Free
LFO 4 Rate	field.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrigger	field.lfo4_retrigger	off / on	off
LFO 4 Fade	field.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	field.mx1_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 1 Target	field.mx1_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 1 Amount	field.mx1_amount	-1 .. 1	0
Mod 2 Source	field.mx2_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 2 Target	field.mx2_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 2 Amount	field.mx2_amount	-1 .. 1	0
Mod 3 Source	field.mx3_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 3 Target	field.mx3_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 3 Amount	field.mx3_amount	-1 .. 1	0
Mod 4 Source	field.mx4_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 4 Target	field.mx4_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 4 Amount	field.mx4_amount	-1 .. 1	0
Mod 5 Source	field.mx5_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 5 Target	field.mx5_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 5 Amount	field.mx5_amount	-1 .. 1	0
Mod 6 Source	field.mx6_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 6 Target	field.mx6_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 6 Amount	field.mx6_amount	-1 .. 1	0
Mod 7 Source	field.mx7_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 7 Target	field.mx7_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off
Mod 7 Amount	field.mx7_amount	-1 .. 1	0
Mod 8 Source	field.mx8_src	Off LFO 1 LFO 2 LFO 3 LFO 4 Mod Env Filter Env Velocity Key Random	Off
Mod 8 Target	field.mx8_dst	Off Pulse Width Cutoff Resonance Filter Mode Level	Off

Mod 8 Amount	field.mx8_amount	-1 .. 1	0
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SPACE & OUTPUT

Control	Key	Range	Default
Width	field.width	0 .. 2	1
Pan	field.pan	-1 .. 1	0
Duck	field.duck	0 .. 1	0.4
Room Send	field.room_send	0 .. 1	0.1
Plate Send	field.plate_send	0 .. 1	0
Hall Send	field.hall_send	0 .. 1	0.2

17. Mixer / Master

A channel strip per part: its pan (the six voices), its sends to the room, the hall and its delay or effects bus, its duck under the kick, the mute ("M") and the fader, beside a meter of what the part puts into the mix -- after fader, trance gate and duck, before the master. The green-to-red bar is the RMS level, the line above it the peak, held for a second and a half; the figure under the meter is that peak in dBFS. The knobs of a strip are the same parameters its synth page shows. Kick and bass have no strip level of their own: their fader is the synth's level.

The fader stays where you put it. Where the composer plays a part at another level -- the level match that holds every track's parts to one loudness, a preset's level correction -- a bright tick beside the fader shows the level that plays, joined to the fader's cap by a line.

Beside the strips the track gain and the sidechain's times, and the master chain: bus compressor, mono bass, a two-times oversampled knee clipper, a lookahead true-peak limiter and a safety clip, metered to ITU-R BS.1770.

Auto Gain renders four four-bar windows of the track through the whole master chain, measures them and offsets the master gain so that the output meets the loudness target. It runs when a track is planned, so it uses the knobs as they stand at that moment.

Depth comes in three planes, and three rooms. The front -- kick, bass and the lead -- is dry, the lead with a little of the near room (A: short, early reflections, a 50 ms pre-delay). The middle -- acid, counter, arp and stab -- sends to the plate (B, the main room: 1.5 s in a drop, 4 s in a breakdown, 25 ms pre-delay). The back -- pads and drone -- sends to the plate as well, more of it, and the plate in turn feeds the hall (C, the far room: ten seconds, 400 Hz .. 3 kHz), which the bed and the voices reach directly. The far room is heard only in intros and breakdowns: in a drop there are only A and B. Pre-delays are distance cues, not tempo values: near sources get the long one (their direct sound arrives well before the first reflection), far ones almost none. Each room has its own send filter (the Rooms group on the FX page).

Every voice carries a Distance (0 near .. 1 far, its own plane by default: the lead 0, the middle 0.5, pad and drone 1). Moving it moves all the cues of distance at once -- level, a low pass, the plate send and the width -- so a voice is never near by one cue and far by another. The composer uses it: the lead steps back in a breakdown and comes forward in the drop, and in the main breakdown the counter-lead arrives from far away over sixteen bars. Pads and drone also move slowly on their own (Slow Movement: cutoff, colour and width on periods of 13, 21 and 34 seconds, in golden-mean ratios, so no pattern is heard to come round).

Every strip gives way to the kick by what its plane needs: the bass 4 dB, the lines about 2 dB and back within 65 ms (Duck Release (Lines)), the back plane 5 to 6 dB within 125 ms (Duck Release), the returns 5 dB within 100 ms (Return Duck Release). The effects do not duck: an impact lands on the drop's kick. The lead drives two ducks of its own: the counter steps back while the lead plays (Counter under Lead), and so does the pad's presence band between 500 Hz and 3 kHz, for as long as a lead note sounds (Pad Band under Lead).

The composer mixes each phase differently. A breakdown stands 7 dB under the drop before it (after the master's compressor and limiter that is about 5 dB of loudness), with the pad at full width and a little louder, the lead further back and the far room open; a buildup narrows the pad, pulls the plate down to 1.2 seconds, fades the far room out and lifts the lines' high pass, and the drop opens it all again within a bar -- the lead at the front, the pad at its full width again, the pad and drone ducking deeper. In the intro and the breakdown, where pad and drone are often the only tonal voices, their unison stays within about six cents: held, a wider stack beats. While the bass plays, the pad leaves the third to the lead. On the last beat before every eight-bar change of a groove or a drop the effects are thrown into the plate for that one beat.

Monitor (Master page): Mono, Sub (a low pass at 80 Hz, where the kick-bass gap and the tuning are judged) or Side, after the meter -- a listening switch, it changes nothing the loudness readings see.



The Mixer / Master tab.

CHANNELS

Control	Key	Range	Default
Level	kick.level	-36 .. 6 dB	-6
Kick Mute	mix.kick_mute	off / on	off
Level	bass.level	-36 .. 6 dB	-3
Bass Mute	mix.bass_mute	off / on	off
Perc Level	mix.perc_level	-24 .. 12 dB	1.2
Perc Mute	mix.perc_mute	off / on	off
Perc Room	mix.perc_room	0 .. 1	0.12
Perc Hall	mix.perc_hall	0 .. 1	0
Acid Level	mix.acid_level	-24 .. 12 dB	0
Acid Mute	mix.acid_mute	off / on	off
Room Send	acid.room_send	0 .. 1	0.1
Hall Send	acid.hall_send	0 .. 1	0
Delay Send	acid.delay_send	0 .. 1	0.25
Duck	acid.duck	0 .. 1	0.15
Lead Level	mix.lead_level	-24 .. 12 dB	-6
Lead Mute	mix.lead_mute	off / on	off
Pan	lead.pan	-1 .. 1	-0.2
Room Send	lead.room_send	0 .. 1	0.1
Hall Send	lead.hall_send	0 .. 1	0
Delay Send	lead.delay_send	0 .. 1	0.3
Duck	lead.duck	0 .. 1	0.2
Counter Level	mix.counter_level	-24 .. 12 dB	5
Counter Mute	mix.counter_mute	off / on	off
Pan	counter.pan	-1 .. 1	0.2
Room Send	counter.room_send	0 .. 1	0
Hall Send	counter.hall_send	0 .. 1	0

Delay Send	counter.delay_send	0 .. 1	0.4
Duck	counter.duck	0 .. 1	0.2
Arp Level	mix.arp_level	-24 .. 12 dB	-5
Arp Mute	mix.arp_mute	off / on	off
Pan	arp.pan	-1 .. 1	0
Room Send	arp.room_send	0 .. 1	0
Hall Send	arp.hall_send	0 .. 1	0
Delay Send	arp.delay_send	0 .. 1	0.35
Duck	arp.duck	0 .. 1	0.2
Stab Level	mix.stab_level	-24 .. 12 dB	-3
Stab Mute	mix.stab_mute	off / on	off
Pan	stab.pan	-1 .. 1	0
Room Send	stab.room_send	0 .. 1	0
Hall Send	stab.hall_send	0 .. 1	0
Delay Send	stab.delay_send	0 .. 1	0.45
Duck	stab.duck	0 .. 1	0.2
Pad Level	mix.pad_level	-24 .. 12 dB	0
Pad Mute	mix.pad_mute	off / on	off
Pan	pad.pan	-1 .. 1	0
Room Send	pad.room_send	0 .. 1	0
Hall Send	pad.hall_send	0 .. 1	0
Delay Send	pad.delay_send	0 .. 1	0
Duck	pad.duck	0 .. 1	0.5
Drone Level	mix.drone_level	-24 .. 12 dB	0
Drone Mute	mix.drone_mute	off / on	off
Pan	drone.pan	-1 .. 1	0
Room Send	drone.room_send	0 .. 1	0
Hall Send	drone.hall_send	0 .. 1	0
Delay Send	drone.delay_send	0 .. 1	0
Duck	drone.duck	0 .. 1	0.45
SFX Level	mix.sfx_level	-24 .. 12 dB	4.5
SFX Mute	mix.sfx_mute	off / on	off
Room Send	sfx.room_send	0 .. 1	0
Hall Send	sfx.hall_send	0 .. 1	0.5
Duck	sfx.duck	0 .. 1	0
Texture Level	mix.texture_level	-24 .. 12 dB	9
Texture Mute	mix.texture_mute	off / on	off
Room Send	texture.room_send	0 .. 1	0.1
Hall Send	texture.hall_send	0 .. 1	0.35
FX Send	texture.fx_send	0 .. 1	0.3
Duck	texture.duck	0 .. 1	0.45
Vocal Level	mix.vocal_level	-24 .. 12 dB	7
Vocal Mute	mix.vocal_mute	off / on	off
Hall Send	vocal.hall_send	0 .. 1	0.3
FX Send	vocal.fx_send	0 .. 1	0.45
Delay Throw	vocal.throw_send	0 .. 1	0.6
Duck	vocal.duck	0 .. 1	0.4
Field Level	mix.field_level	-24 .. 12 dB	-14
Field Mute	mix.field_mute	off / on	off
Pan	field.pan	-1 .. 1	0
Room Send	field.room_send	0 .. 1	0.1
Hall Send	field.hall_send	0 .. 1	0.2
Duck	field.duck	0 .. 1	0.4

TRACK & SIDECCHAIN

Control	Key	Range	Default
Track Gain	mix.track_gain	-12 .. 12 dB	0

Duck Attack	<code>mix.duck_attack</code>	0.5 .. 30 ms	2
Duck Hold	<code>mix.duck_hold</code>	0 .. 200 ms	20
Duck Release	<code>mix.duck_release</code>	10 .. 800 ms	125
Duck Release (Lines)	<code>mix.duck_release_lines</code>	10 .. 800 ms	65
Counter under Lead	<code>mix.counter_duck</code>	0 .. 1	0.2
Pad Band under Lead	<code>mix.pad_lead_duck</code>	0 .. 12 dB	3

GAIN

Control	Key	Range	Default
Gain	<code>master.gain</code>	-24 .. 24 dB	0
Ceiling	<code>master.ceiling</code>	-12 .. 0 dBTP	-1
Clip	<code>master.clip</code>	off / on	on

COMPRESSOR

Control	Key	Range	Default
Comp Threshold	<code>master.comp_threshold</code>	-40 .. 0 dB	-8
Comp Ratio	<code>master.comp_ratio</code>	1 .. 10:1	1.5
Comp Knee	<code>master.comp_knee</code>	0 .. 24 dB	6
Comp Attack	<code>master.comp_attack</code>	0.1 .. 100 ms	30
Comp Release	<code>master.comp_release</code>	10 .. 2000 ms	200

OUTPUT

Control	Key	Range	Default
Mono Bass	<code>master.mono_bass</code>	40 .. 300 Hz	120
Limiter	<code>master.limiter</code>	off / on	on
Limiter Release	<code>master.limiter_release</code>	2 .. 1000 ms	20
Target Loudness	<code>master.target_lufs</code>	-20 .. -4 LUFS	-9
Auto Gain	<code>master.auto_gain</code>	off / on	on
Clipper	<code>master.clipper</code>	off / on	on
Clipper Threshold	<code>master.clipper_threshold</code>	-6 .. 6 dB	0

MONITOR

Control	Key	Range	Default
Monitor	<code>master.monitor</code>	Normal Mono Sub Side	Normal

18. Perform

Four macros, meant for a hand rather than a mouse.

They are immediate parameter moves rather than score events, and that is measured rather than convenient: the engine's control ring is first in, first out and the composer keeps it filled eight bars ahead, so an event pushed into it now would not be heard for about thirteen seconds. A macro therefore writes the knob itself, as an offset in the knob's normalised domain, and puts it back exactly where it was when you let go. The composer's own ramps keep riding on top.

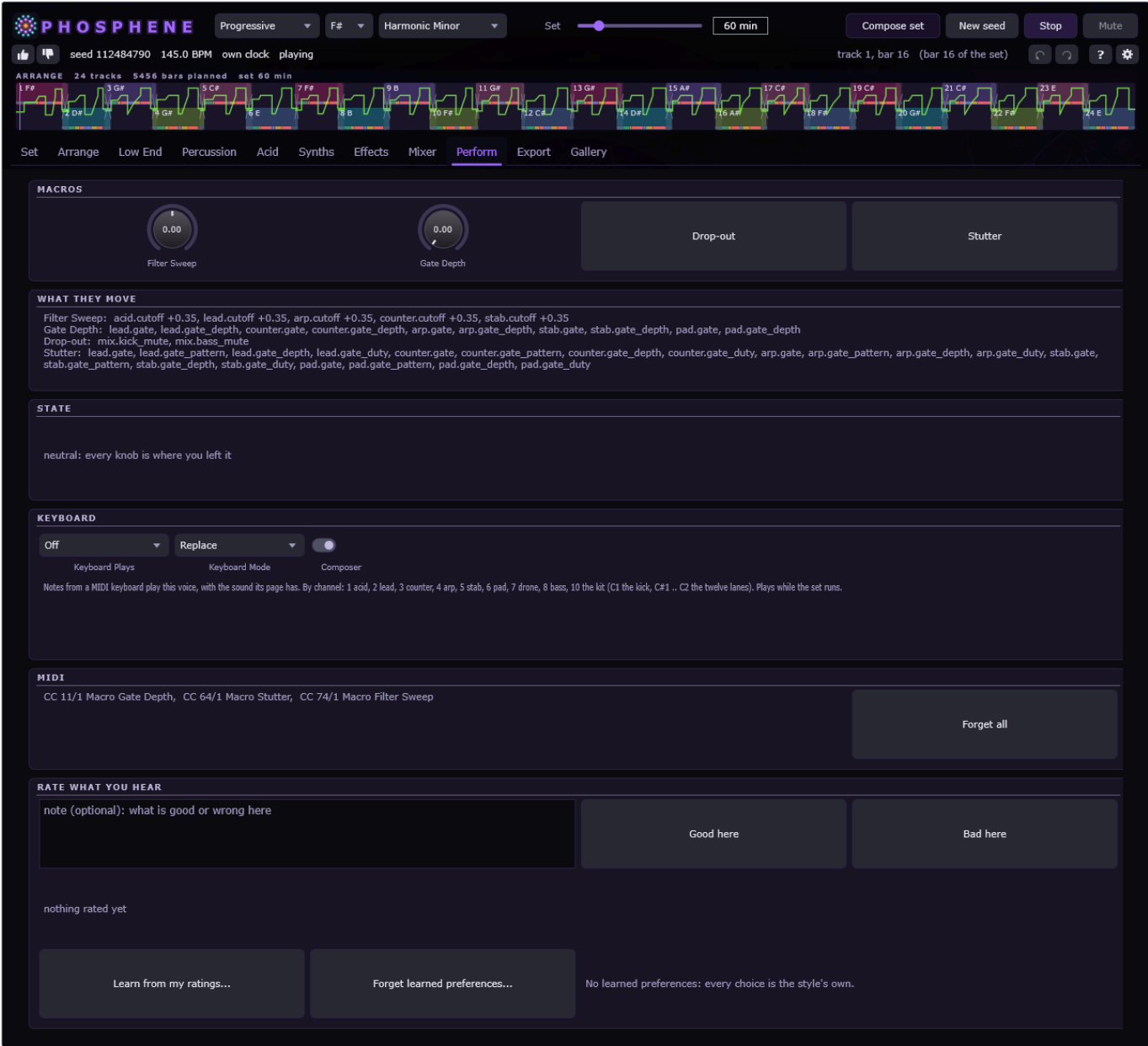
Drop-out is a press, not a hold: it takes the kick and the bass out and lets them back in on the next bar line. Stutter is held. Filter Sweep and Gate Depth stay where you leave them, and a double click puts them back to neutral.

Keyboard: notes from a MIDI keyboard can play one voice -- the acid, any of the six melodic voices, the bass or the kit (C1 the kick, C#1 to C2 the twelve lanes), or "By channel" (channel 1 the acid, 2 the lead ... 7 the drone, 8 the bass, 10 the kit) -- with the sound that voice's tab has. Replace leaves the generator's notes of that voice out, Layer plays over them. Composer off leaves every generated note out: then only what you play sounds, through the mix and the effects as composed. The keyboard plays while the set runs; an export always plays what was composed.

MIDI: every knob, switch and chooser, and the four macros, can be put on a hardware controller. Right click the control, choose MIDI Learn and turn the knob on your controller. The MIDI group lists every binding and forgets them all at once. Without learning, controller 74 is the filter sweep, 11 the gate depth and 64 the stutter -- the controllers every generator of the family listens to.

Headset: while a Meta Quest sends its hands (the Quest app's bridge mode), the Perform tab shows what they do -- the left pinch play and stop, the right pinch the drop-out, held the stutter, both hands the next track, the left hand the filter sweep, the right the gate depth (The panel).

Rate what you hear: Good here and Bad here write a verdict with a note into your ratings. Learn from my ratings fits your taste to them and, once you confirm, weights the choices the style leaves open towards what you liked. The genre's rules always stay above your taste.



The Perform tab.

Filter Sweep

Opens or closes the filters of acid, lead, counter-lead, arp and stab together, by up to a third of their range in each direction. Centre is neutral.

Moves: acid.cutoff +0.35, lead.cutoff +0.35, arp.cutoff +0.35, counter.cutoff +0.35, stab.cutoff +0.35

Gate Depth

Switches the trance gate on for lead, counter-lead, arp, stab and pad and sets its depth; at 1.0 the gate closes completely between its steps.

Moves: lead.gate = 1, lead.gate_depth = 1, counter.gate = 1, counter.gate_depth = 1, arp.gate = 1, arp.gate_depth = 1, stab.gate = 1, stab.gate_depth = 1, pad.gate = 1, pad.gate_depth = 1

Drop-out (press)

Takes kick and bass out until the next bar line and lets them back in on the downbeat. One press, one bar.

Moves: mix.kick_mute = 1, mix.bass_mute = 1

Stutter (press)

Held: lead, counter-lead, arp, stab and pad run through the gate on sixteenths at full depth and the shortest duty. Phosphene has no buffer repeat, so this is the gate's stutter, not a tape one.

Moves: lead.gate = 1, lead.gate_pattern = 0, lead.gate_depth = 1, lead.gate_duty = 0, counter.gate = 1, counter.gate_pattern = 0, counter.gate_depth = 1, counter.gate_duty = 0, arp.gate = 1, arp.gate_pattern = 0, arp.gate_depth = 1, arp.gate_duty = 0, stab.gate = 1, stab.gate_pattern = 0, stab.gate_depth = 1, stab.gate_duty = 0, pad.gate = 1, pad.gate_pattern = 0, pad.gate_depth = 1, pad.gate_duty = 0

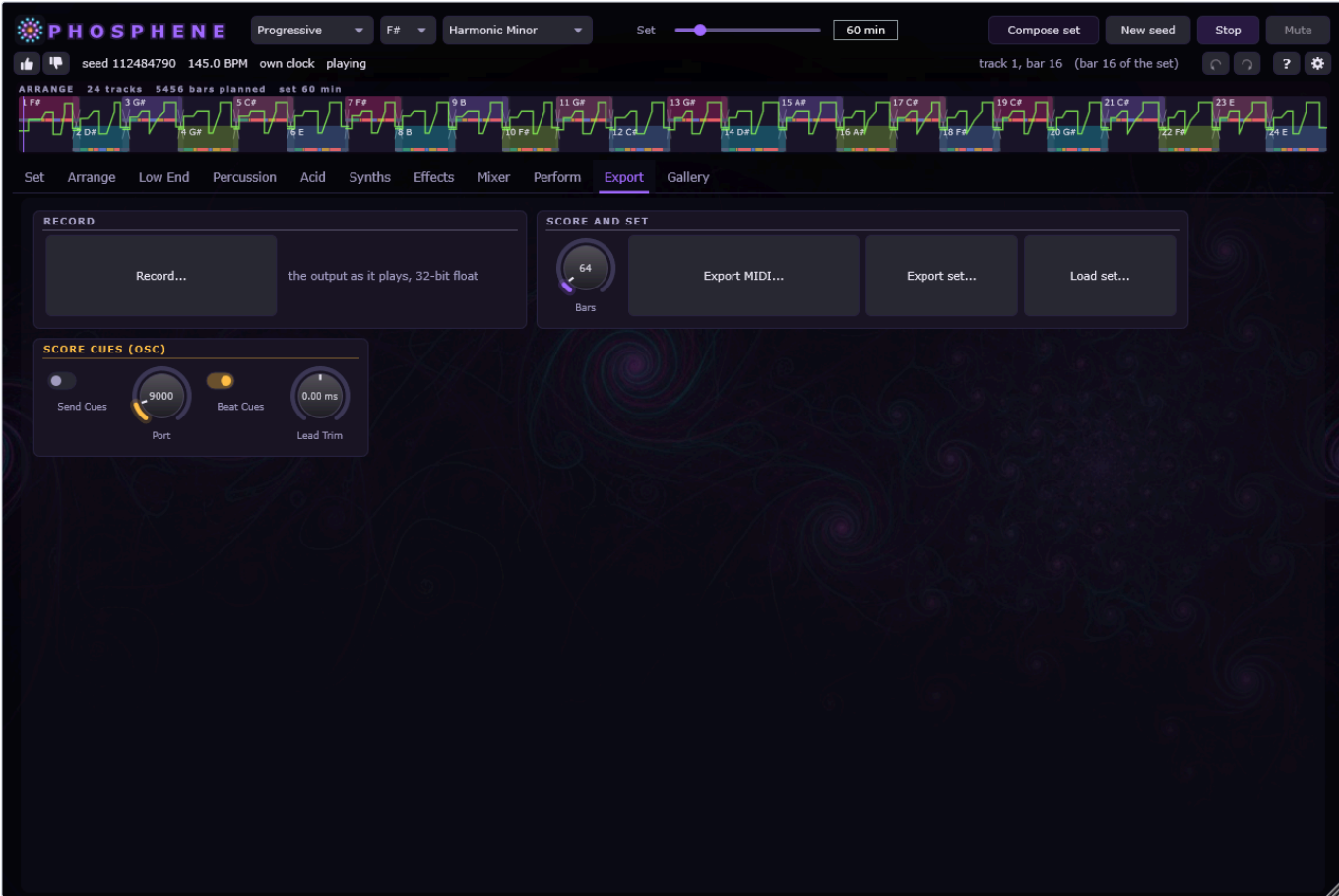
KEYBOARD

Control	Key	Range	Default
Keyboard Plays	mix.keyboard_part	Off Acid Lead Counter Arp Stab Pad Drone By channel Bass Kit	Off
Keyboard Mode	mix.keyboard_mode	Replace Layer	Replace
Composer	mix.composer	off / on	on

19. Export

What leaves the instrument, in one place, as every generator of the family has it (01.10.2026): Record writes the output as it plays into a WAV file until it is pressed again; Export MIDI writes the score of so many bars from the start, a channel per part; Export set writes the set -- the seed, the locks, the rerolls and every knob that differs from the factory -- as a .phosset, and Load set reads one and plays it. The score cues (OSC) send the set's beats, bars, sections and keys to a light show or a VJ program at the moment they are heard.

The keys: Ctrl+E exports the MIDI, Ctrl+S the set, Ctrl+O loads one.



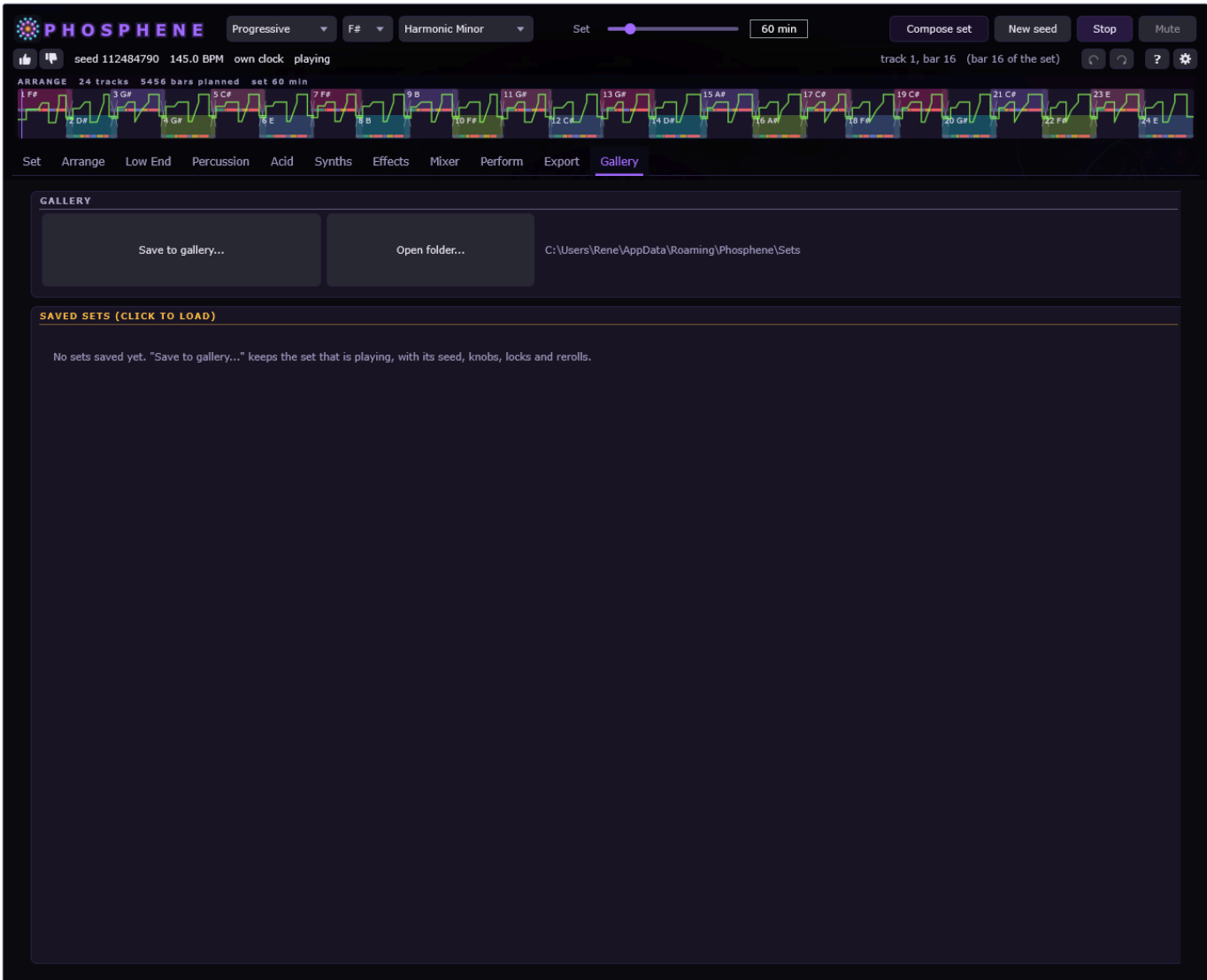
The Export tab.

SCORE CUES (OSC)

Control	Key	Range	Default
Send Cues	cue.send	off / on	off
Port	cue.port	1024 .. 65535	9000
Beat Cues	cue.beats	off / on	on
Lead Trim	cue.lead_ms	-50 .. 50 ms	0

20. Gallery

Your saved sets, each with a picture of its form. Save to gallery keeps the set as it stands -- seed, style, arc, knobs, locks and rerolls -- and a click on an entry loads it back. The files are ordinary .phosset files in your Sets folder; Open folder shows it.



The Gallery tab.

21. Filters and modulation

The melodic voices and the bass choose their filter's circuit with Filter Model. The first entry is the synth's own filter -- the voices' state-variable filter, the bass's ladder -- and the others are circuit models solved sample by sample: the Moog transistor ladder, the OTA cascades of the Prophet and the Juno, the Oberheim SEM, the Xpander's pole mixes, the diode ladder of the TB-303, the Korg35 of the MS-20, the Polivoks and the Wasp. Filter Mode morphs the SEM from low pass through notch to high pass, steps through the Xpander's eight responses and turns the Polivoks into a band pass. Switching models keeps the level: each is calibrated against the synth's own filter. The acid keeps its diode ladder -- that is the acid.

The filter envelope is a full ADSR: Filter Attack, Decay, Sustain and Release. With the attack at its minimum and the sustain at 0 it is the plain decay every older set used.

A third envelope (Mod Envelope), four LFOs and a modulation matrix of eight slots move the rest. An LFO runs free at its rate or synced to the beat (4 bars down to a sixteenth, and triplets), in seven shapes from sine to smooth random; Retrigger restarts it with every note, Fade brings it in over a few seconds. A slot of the matrix reads a source -- an LFO, the mod or filter envelope, the velocity, the key or a random value drawn per note -- and moves a target by its amount: pitch, the second oscillator's pitch, pulse width, the wavetable position (the classic slow scan through a table), the FM index, cutoff, resonance, the filter mode, the level or the pan. The bass offers only what cannot break its lock to the kick (no pitch, no pan), the acid what suits a 303 line.

22. Sound presets

Every synth tab starts with a Sound group: the synth's presets in a chooser, grouped by kind, a Save... button for presets of your own, and the Own Sound switch.

Every track of a set plays a preset on every synth that is not on Own Sound: the composer chooses one by the track's style -- a Hoover or a Supersaw Anthem for Full-On, a Growl or a Screaming Glide for Dark Forest, a Glass Lead or a Psy Pluck for Progressive -- and never the same group as the two tracks before. While it plays, the tab shows it: the chooser reads "Group: Name (composer)" and the knobs stand at the preset's values. The sections still move some of them on top, the way a player rides a filter; those moves are not written into the knobs. The composer also evens out the presets' loudness: each carries a correction, measured against the synth's own default sound, that rides on the synth's Level on top of your setting, and the level match then holds every part of every track -- the first one included -- to the loudness the default sound would have there. Sound Presets on the Set page (or Sound Variation at 0) turns the composer's presets off, and every synth plays its knobs again, with the composer's recipes.

Own Sound keeps the composer out: the synth then sounds exactly as its tab says, in every track. Turning a knob of a synth that plays the composer's preset switches its Own Sound on, with the knobs where the preset put them, so the sound you start from is the one you heard. Choosing a preset from the chooser switches Own Sound on as well. A preset sets the synth's sound and leaves its level, its ducking, its sends to the rooms, its place in the image and -- on the melodic voices -- its high pass and trance gate alone, since those are where the mix and the form put the voice. Undo takes a preset back.

The live ring: where a knob plays away from where it stands -- a section riding the cutoff, drop 2's lift, a preset's level correction -- a thin bright arc on the knob's ring runs from the knob to the value that plays, with a dot where it plays. The modulation shows there too: an LFO or the mod envelope on the cutoff, the table position, the resonance, the filter mode, the pulse width, the FM index, the level or the pan moves the dot as the latest note's voice moves. A knob without it plays exactly what it shows.

The factory bank holds 1024 presets for each synth: sixteen groups -- Hoover, Acid Lead, Vocal Formant, Rolling, Squelchy, Classic 303, Full-On ... -- of sixty-four each. A group is a grid of eight adjectives from dark to bright (the lead's run from Umbral to Supernova) and eight nouns of its own (the Hoover's from Hoover to Titan), so "Umbral Mentasm" is the darkest of the Hoovers' second column and "Supernova Titan" the brightest of its last. Each preset brings the filter model its group suits and modulation from its group's recipes: a slow LFO on the cutoff, the wavetable scanned by an LFO or the mod envelope, a hoover's pitch dive, a vibrato that fades in. Your own presets are listed under User and live in the Presets folder of your Phosphene user folder.

23. The panel

Phosphene shares its panel with the other generators of the family -- Ephemeris, Totality and Parhelion (and, in part, Noctuary) -- so that what is learned on one holds on the others: the same places, the same keys, the same controllers, the same hands. Each keeps its own colours, its own letters and the picture behind its panel (Settings > Picture behind the panel takes it away).

The header's first row: the logo, the style, the key and the scale, what is composed and its length, Compose, New seed, Play and Mute. Its second row: Good here and Bad here -- a verdict on what plays, for your taste --, the status line, and at the right Undo, Redo, Help and the settings. Under the header the whole set at a glance; a click jumps there. The tabs come in one order everywhere: Set, Arrange, the instrument's own pages -- a page of several has small tabs of its own --, Mixer, Perform, Export, Gallery.

A page taller than the window shows its groups in sections, one at a time, with a switch at its top right: a synth's sound and its modulation apart (Sound, Modulation), and where even those do not fit, cut further, each named after its first group (Filter, LFO, Matrix ...). At the window's usual size, 1280 by 860, no page scrolls. The overview above the tabs can be folded away in the settings; the pages then get its height.

A colour says what a control does, the same on every page and in every generator of the family: the sources, the filters, the envelopes, what moves by itself, the room and the mix. A knob the composer moves has a thin ring round it that shows where it stands at this moment. A double click sets a control to its default; a right click offers MIDI learn, forgetting the controller, and the default.

Undo and Redo take back and bring back what was changed -- a knob, a new seed, a reroll, a preset, a loaded set -- one step at a time; the button's tooltip names the step. Help opens this manual by topic, with the page in front, the keys and the headset as topics of their own. The settings hold the update check (it asks GitHub once a day whether a newer Phosphene is out; nothing else is sent), the picture behind the panel, the overview, the headset, the window's size, full screen, the keys and About.

Space	play / stop
Ctrl+Z	undo
Ctrl+Y, Ctrl+Shift+Z	redo
F1	help
F11	full screen (the standalone); Esc leaves it and closes the help
Ctrl+S, Ctrl+O	save the set, open one
Ctrl+E	export

The controllers every generator of the family listens to without learning: controller 74 (brightness) is the filter sweep, 11 (the expression pedal) the gate depth, 64 (the sustain pedal) the stutter while it is held; the mod wheel is left free. A set with bindings of its own keeps them.

The headset. The controls of a Meta Quest's hands are shown only while a headset sends them -- or always, if the settings say so (Headset > Always show them); Off stops listening. The Quest app sends its hands to this computer in bridge mode: bridge_host = this computer's address in its phos.cfg (audio=0 leaves the headset silent), and Phosphene listens on port 9101. The hands mean the same in every generator of the family, on the headset alone and here:

left pinch	play / stop
both hands pinched	the next track
right pinch	drop-out: kick and bass gone until the next bar line
right pinch held	stutter, while it is held
left hand up, down	filter sweep: closed below the middle, open above
right hand up	gate depth

A pinch acts when it opens again, so a pinch of both hands is never also one of each; a hand moves its control only while it does not pinch, and both glide and are centred: a hand at mid height plays what was composed.

24. Shortcuts

Space	play / stop
F1	help (this page); Esc closes it
F11	full screen (standalone)
Ctrl+Z	undo
Ctrl+Y	redo (also Ctrl+Shift+Z)
Ctrl+S, Ctrl+O	export the set, load one
Ctrl+E	export MIDI
right click	MIDI Learn / Forget MIDI / Default value on any control
double click	a knob back to its default

The window can be dragged to any size; the whole panel zooms with it. The standalone's title bar has a maximise button, and Full screen in the settings (or F11) fills the screen.

25. Updates

Phosphene looks for a newer release at most once a day, when its window opens, by asking GitHub for the latest release of the project -- nothing else is sent, and nothing is installed by itself. When there is one, the header says so and a click opens the release page. The check can be switched off here on the help page or in the settings, and Check now asks at once.

Phosphene 1.2.0 · generated from the plugin: the parameter tables and the pictures come out of the running editor (Tools/manual/make_manual.py).